



Bystander IFN- γ activity promotes widespread and sustained cytokine signaling altering the tumor microenvironment

Ronan Thibaut^{1,2}, Pierre Bost^{3,4,5,6}, Idan Milo^{1,6}, Marine Cazaux^{1,2}, Fabrice Lemaître¹, Zacarias Garcia¹, Ido Amit³, Béatrice Breart¹, Clémence Cornuot¹, Benno Schwikowski⁴ and Philippe Bousso¹✉

The cytokine interferon (IFN)- γ produced by tumor-reactive T cells is a key effector molecule with pleiotropic effects during anti-tumor immune responses. Although IFN- γ production is targeted at the immunologic synapse, its spatiotemporal activity within the tumor remains elusive. In the present study, we report that, although IFN- γ secretion requires local antigen recognition, IFN- γ diffuses extensively to alter the tumor microenvironment in distant areas. Using intravital imaging and a reporter for STAT1 translocation, we provide evidence that T cells mediate sustained IFN- γ signaling in remote tumor cells. Furthermore, tumor phenotypic alterations required several hours of exposure to IFN- γ , a feature that disfavored local IFN- γ activity over diffusion and bystander activity. Finally, single-cell RNA-sequencing data from melanoma patients also suggested bystander IFN- γ activity in human tumors. Thus, tumor-reactive T cells act collectively to create large cytokine fields that profoundly modify the tumor microenvironment.

IFN- γ is a key soluble effector molecule during anti-tumor immune responses. Mice deficient for IFN- γ production are prone to develop spontaneous or carcinogen-induced tumors, highlighting the crucial importance of IFN- γ during immunosurveillance^{1–3}. IFN- γ also plays an important role during tumor immunotherapy because the benefit of anti-programmed death 1 (anti-PD-1) therapy in mice has been shown to be dependent on IFN- γ production by CD8⁺ T cells and the associated interleukin (IL)-12 production by dendritic cells⁴. Moreover, in melanoma patients, acquired resistance to anti-CTLA-4 therapy has been associated with mutations in the IFN- γ signaling pathway⁵.

Several mechanisms have been described to explain the beneficial impact of IFN- γ in the tumor microenvironment. For example, IFN- γ can exert direct cytotoxic or cytostatic effects on tumor cells^{6,7}, and contribute to tumor senescence⁸ and tumor ferroptosis⁹. In addition, IFN- γ signaling on stromal cells can lead to reduced angiogenesis in the tumor bed, limiting nutrients and dioxygen access^{7,10}. IFN- γ signaling can also lead to major histocompatibility complex (MHC) class I upregulation, thereby increasing tumor sensitivity to CD8⁺ T cell-mediated lysis¹¹. Finally, IFN- γ orchestrates the recruitment of natural killer (NK) cells, T cells and invariant NK T cells to the tumor by triggering production of the chemokines CXCL9, CXCL10 and CXCL11 (ref. ¹²).

Conversely, intratumoral IFN- γ may have detrimental effects on the outcome of anti-tumor immune responses. In this regard, it has been reported that IFN- γ increases tumor cell genomic instability¹³, a process that may favor tumor escape from the immune pressure. IFN- γ can also induce the expression of ligands for inhibitory receptors such as PD ligand 1 (PD-L1) and PD-L2 on stromal and tumor cells¹⁴, contributing to the inhibition of tumor-infiltrating T cells^{15,16}. IFN- γ could also induce the deletion of tumor-reactive T cells by

favoring activation-induced cell death¹⁷. Finally, long-term IFN- γ exposure has been associated with deep transcriptomic changes in tumor cells that can contribute to the failure of immune checkpoint blockade therapy¹⁸.

Despite our understanding of molecular and cellular mechanisms triggered by IFN- γ , we critically lack information on the spatiotemporal activity of this cytokine in the tumor microenvironment. Quantitative rules regulating cell–cell communication are only starting to be considered¹⁹ but are essential to understand complex immune responses. Fundamental questions remain to be addressed. Does IFN- γ act primarily in discrete areas or more widely in the tumor microenvironment? How is IFN- γ spatial activity shaped by the number of cytokine-producing T cells? What is the duration of IFN- γ exposure required to alter tumor cell function and phenotype? It is well established that IFN- γ is secreted by T cells in a directional manner toward the immunologic synapse formed with target cells²⁰. On T cell receptor (TCR) stimulation, IFN- γ production is also thought to be a brief (few hours' long) event^{21,22}. Although these two features are expected to favor a local activity of IFN- γ around the sites of cytokine production, the extent of IFN- γ diffusion in tissue has yet to be fully understood. Indeed, using in vitro settings, it has been shown that the immunologic synapse does not perfectly restrict IFN- γ secretion, which accounts for bystander activity of IFN- γ in cells that are not in contact with cytokine-producing T cells²³. Evidence for bystander IFN- γ activity has also been reported in vivo in the mesenteric lymph node after *Toxoplasma gondii* infection²⁴ and in the dermis infected with *Leishmania major*²⁵. Yet, the extent and functional consequences of IFN- γ diffusion in the microenvironment of distinct tumor types remain unclear.

In the present study, we investigated the spatiotemporal activity of IFN- γ in the tumor microenvironment using B cell lymphoma

¹Dynamics of Immune Responses Unit, Equipe Labellisée Ligue Contre le Cancer, Institut Pasteur, INSERM U1223, Paris, France. ²University Paris Diderot, Sorbonne Paris Cité, Cellule Pasteur, Paris, France. ³Department of Immunology, Weizmann Institute of Science, Rehovot, Israel. ⁴Systems Biology Group, Center for Bioinformatics, Biostatistics, and Integrative Biology and USR 3756, Institut Pasteur CNRS, Paris, France. ⁵Sorbonne Université, Paris, France.

⁶These authors contributed equally: Pierre Bost, Idan Milo. ✉e-mail: philippe.bousso@pasteur.fr

and melanoma models. Using distinct approaches including mosaic tumors and intravital imaging with a real-time reporter of signal transducer and activator of transcription 1 (STAT1) nuclear translocation, we show that IFN- γ signaling occurs with extensive bystander activity throughout the tumor microenvironment, affecting both tumor and tumor-infiltrating immune cells. Although STAT1 activity in tumor cells was sustained, it did not require continuous interaction with T cells, confirming the extensive bystander activity of intratumoral IFN- γ . We provide evidence that prolonged IFN- γ exposure is in fact needed to alter tumor cell phenotype. Our results support a model in which IFN- γ regulates the tumor microenvironment by acting as a widespread and sustained cytokine field, which concentration is dependent on the collective T cell activity.

Results

T cell-derived IFN- γ profoundly modifies the tumor microenvironment. To characterize how intratumoral IFN- γ acts on tumor cells, we examined the expression of several markers, including MHC class I and PD-L1, in distinct tumor cell lines, on exposure to various concentrations of IFN- γ . Using models for Myc-driven B cell lymphoma (E μ -myc)²⁶, melanoma (B16.F10)²⁷ and mammary tumor (E0771)²⁸, we observed a dose-dependent upregulation of these markers for all cell lines (Fig. 1a,b, and see Extended data Fig. 1a). Phenotypic changes on IFN- γ treatment followed an analog, rather than a digital, pattern with a concentration-dependent shift in expression of the whole population rather than a bimodal distribution (see Extended data Fig. 1b). At high concentrations, IFN- γ also induced substantial cell death of B16.F10 tumor cells and an even more pronounced effect when combined with tumor necrosis factor (TNF)- α (see Extended data Fig. 1c,e). By contrast, IFN- γ did not induce substantial cell death of E μ -myc cells either alone or in combination with TNF- α , highlighting the diversity of the tumor response to this cytokine (see Extended data Fig. 1d,f).

To test whether these phenotypic changes also occur in response to intratumoral T cell-derived IFN- γ in vivo, we adoptively transferred OT-I CD8⁺ T cells into *Rag2*^{-/-} mice bearing ovalbumin⁺ (OVA⁺) E μ -myc lymphomas and analyzed tumor phenotype in the bone marrow, a primary site of tumor growth (Fig. 1c, and see Extended data Fig. 2). We used recipients devoid of endogenous T cells for better control of the source of IFN- γ production. A fraction of OT-I T cells at the tumor site was found to produce IFN- γ (Fig. 1d). As shown in Fig. 1e,f, the presence of T cells resulted in an increased expression of MHC class I and PD-L1 on tumor cells. Phenotypic changes were also detected in tumor-infiltrating immune cells that expressed increased levels of MHC class I on their surface (Fig. 1g). All these effects appeared to be dependent

on T cell-derived IFN- γ because they were abolished when IFN- γ -deficient OT-I T cells were transferred. Importantly, these results were recapitulated in *Rag2*^{-/-} γ ^{-/-} recipients that lack NK cells, suggesting that T cell-derived IFN- γ is sufficient to directly mediate phenotypic changes in tumor cells (Extended data Fig. 3). To extend these findings in a setting of chimeric antigen receptor (CAR) T cell therapy, we analyzed the phenotypic changes in lymphoma cells on transfer of IFN- γ either sufficient or deficient in anti-CD19 CAR T cells. The presence of wild-type (WT) but not IFN- γ -deficient CAR T cells resulted in a substantial increase in MHC class I and PD-L1 expression (Fig. 1h). Altogether, these results indicate that IFN- γ produced by T or CAR T cells has a broad impact on tumor and tumor-infiltrating immune cells.

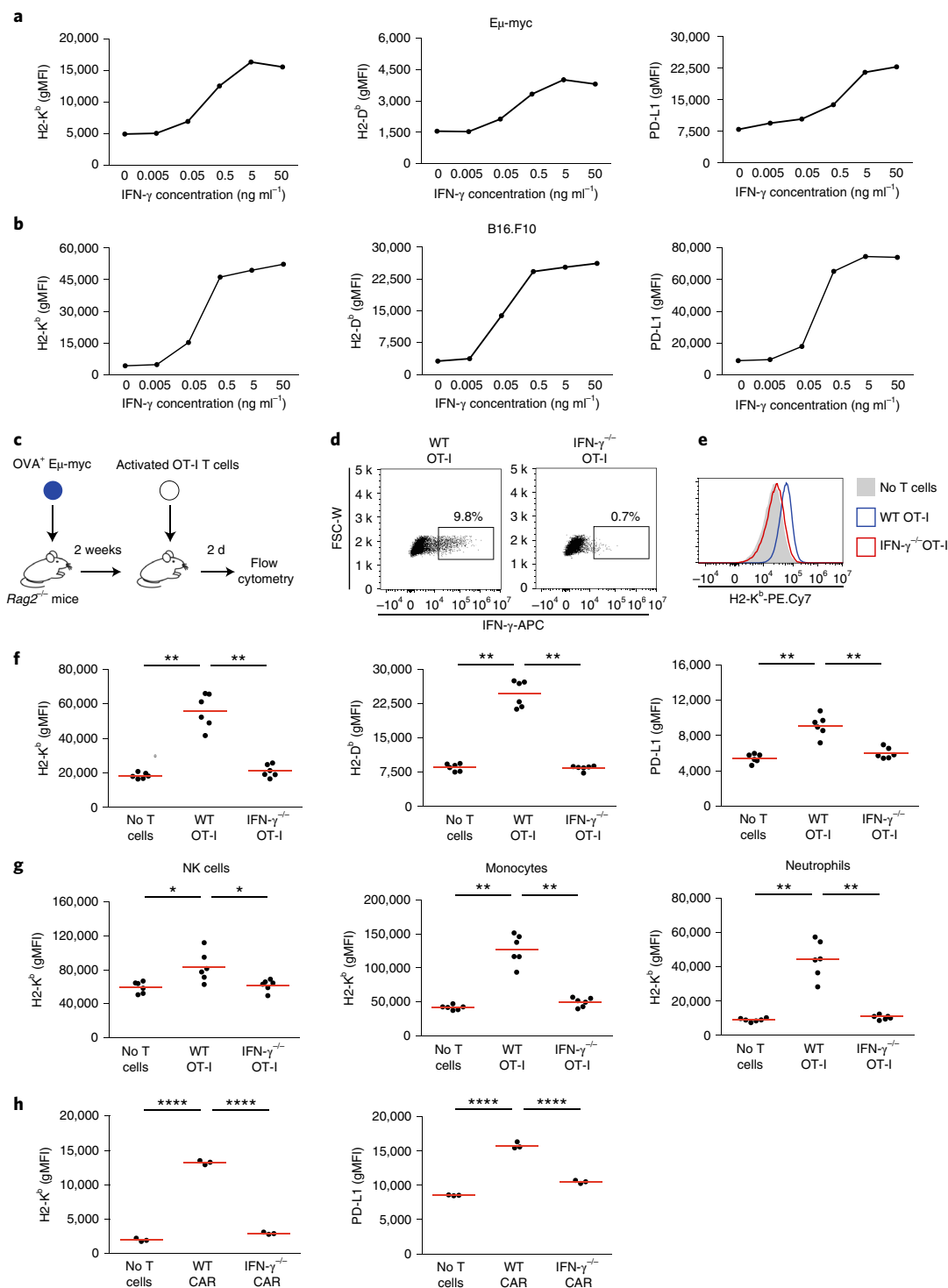
Tumor-infiltrating T cells selectively accumulate and arrest in tumor areas with cognate antigen expression. Production of IFN- γ by T cells has been associated with stable interactions with antigen-presenting targets²². However, it is unclear whether IFN- γ produced by T cells acts on engaged targets or their close neighbors, or more extensively in the tumor microenvironment. To address this question, we wanted to create mosaic tumors in which only selective regions express a tumor antigen to spatially restrict IFN- γ production (Fig. 2a). We took advantage of our previous observation that tumor subclones tend to grow as anatomically segregated patches to create these disparate environments²⁹. When we injected a mixture of antigen-positive (OVA⁺) and antigen-negative E μ -myc tumors expressing distinct fluorescent proteins, we observed tumor patches with a typical radius of 100–500 μ m in the bone marrow (Fig. 2b). Using intravital imaging, we found that most OT-I T cells selectively accumulated and arrested in antigen-positive areas where they displayed reduced velocity and increased confinement, and formed long-lasting interactions with tumor cells (Fig. 2b–g, and see Extended data Fig. 4 and Supplementary Video 1). The remaining T cells seen in antigen-negative tumor areas were rapidly migrating with no evidence for prolonged interactions (Fig. 2b–g, and see Extended data Fig. 4 and Supplementary Video 1). Some of the stable contacts with antigen-positive tumors most probably represent sites of cytokine secretion because previous studies have linked T cell arrest to IFN- γ secretion²². Even if it remains formally possible that migrating T cells also release small amounts of cytokine, the increased T cell numbers and residence time in antigen-positive areas would imply higher cytokine release in these regions. Of note, T cell arrest in antigen-positive areas was largely IFN- γ independent because prolonged interactions were also observed with IFN- γ -deficient T cells (see Extended data Fig. 4). In sum, this strategy provides a means to spatially control cognate interactions

Fig. 1 | T cell-derived IFN- γ induces phenotypic changes in the tumor microenvironment. a,b, IFN- γ induces phenotypic changes in tumor cells in vitro. E μ -myc B lymphoma (a) or B16.F10 melanoma (b) cells were stimulated with indicated IFN- γ concentrations in vitro for 24 h. H2-K^b (left), H2-D^b (middle) and PD-L1 (right) surface expression was then analyzed by flow cytometry. Each dot represents the mean of three technical replicates, representing two (B16.F10) or three (E μ -myc) independent experiments; gMFI, geometric mean fluorescence intensity. c–g, T cell-derived IFN- γ increases MHC class I and PD-L1 levels in tumor and tumor-infiltrating immune cells. c, Experimental setup. *Rag2*^{-/-} mice were injected i.v. with OVA-expressing E μ -myc B lymphoma cells. On days 12–13, in vitro activated WT or IFN- γ -deficient OT-I CD8⁺ T cells were injected i.v. After 2 d, the recipient bone marrow was harvested and analyzed by flow cytometry. d, Intracellular IFN- γ staining was performed in the absence of in vitro restimulation. Dot plots showing IFN- γ production by intratumoral WT, but not IFN- γ -deficient, OT-I T cells (data represent $n = 6$ mice per group). e, Representative example of histograms showing H2-K^b surface expression on tumor cells isolated from the bone marrow of mice that were left untreated (filled gray) or injected with either WT OT-I T cells (line, blue) or IFN- γ -deficient OT-I T cells (line, red). Data represent two independent experiments with $n = 6$ mice per group. f, H2-K^b (left), H2-D^b (middle) and PD-L1 (right) surface expression on tumor cells isolated from mice either treated or not treated with the indicated OT-I T cell population, as assessed by flow cytometry. Each dot represents one mouse. Red lines indicate mean values. Data represent two independent experiments with $n = 6$ mice per group (** $P < 0.01$, two-tailed Mann-Whitney U -test). g, H2-K^b surface expression on NK cells (left), monocytes (middle) and neutrophils (right) isolated from mice either treated or not treated with the indicated OT-I T cells, as assessed by flow cytometry. Red lines indicate mean values. Data represent two independent experiments with $n = 6$ mice per group (* $P < 0.05$, ** $P < 0.01$, two-tailed Mann-Whitney U -tests). h, Recipients with established E μ -myc B cell lymphoma were treated with anti-CD19 CAR T cells and analyzed 2 d later. H2-K^b (left) and PD-L1 (right) surface expression on tumor cells isolated from mice left untreated or treated with WT or IFN- γ -deficient CAR T cells. Red lines indicate mean values. Data represent two independent experiments with $n = 3$ mice per group (**** $P < 0.0001$ using ANOVA, with Tukey's test for multiple comparisons).

leading to IFN- γ production by T cells and to assess the occurrence of long-distance effects in antigen-negative areas.

Strong bystander activity of T cell-derived IFN- γ in the tumor microenvironment. Using the aforementioned setup, we asked how T cells impact on the phenotype of tumor cells located in antigen-positive versus antigen-negative areas. As IFN- γ induces MHC class I and PD-L1 upregulation, we relied on these readouts as a proxy for IFN- γ activity. We first used the E μ -myc lymphoma model. When OT-I T cells were transferred in mice bearing antigen-negative tumors only, no changes in MHC class I and PD-L1

levels were observed (Fig. 3b–d) despite the presence of T cells in the tumor microenvironment (see Extended data Fig. 4e). This confirmed that local antigen restimulation is required for IFN- γ production. In mosaic tumors, the presence of T cells resulted in an increased expression of MHC class I and PD-L1. Importantly, both antigen-positive and antigen-negative tumor cells upregulated these markers to the same extent (Fig. 3a–d). The widespread effects mediated by IFN- γ could be due to the activity of numerous T cells, each of them acting on a limited number of bystander cells (Fig. 3e). Alternatively, efficient diffusion of IFN- γ may result in a relatively uniform intratumoral cytokine concentration, lead-



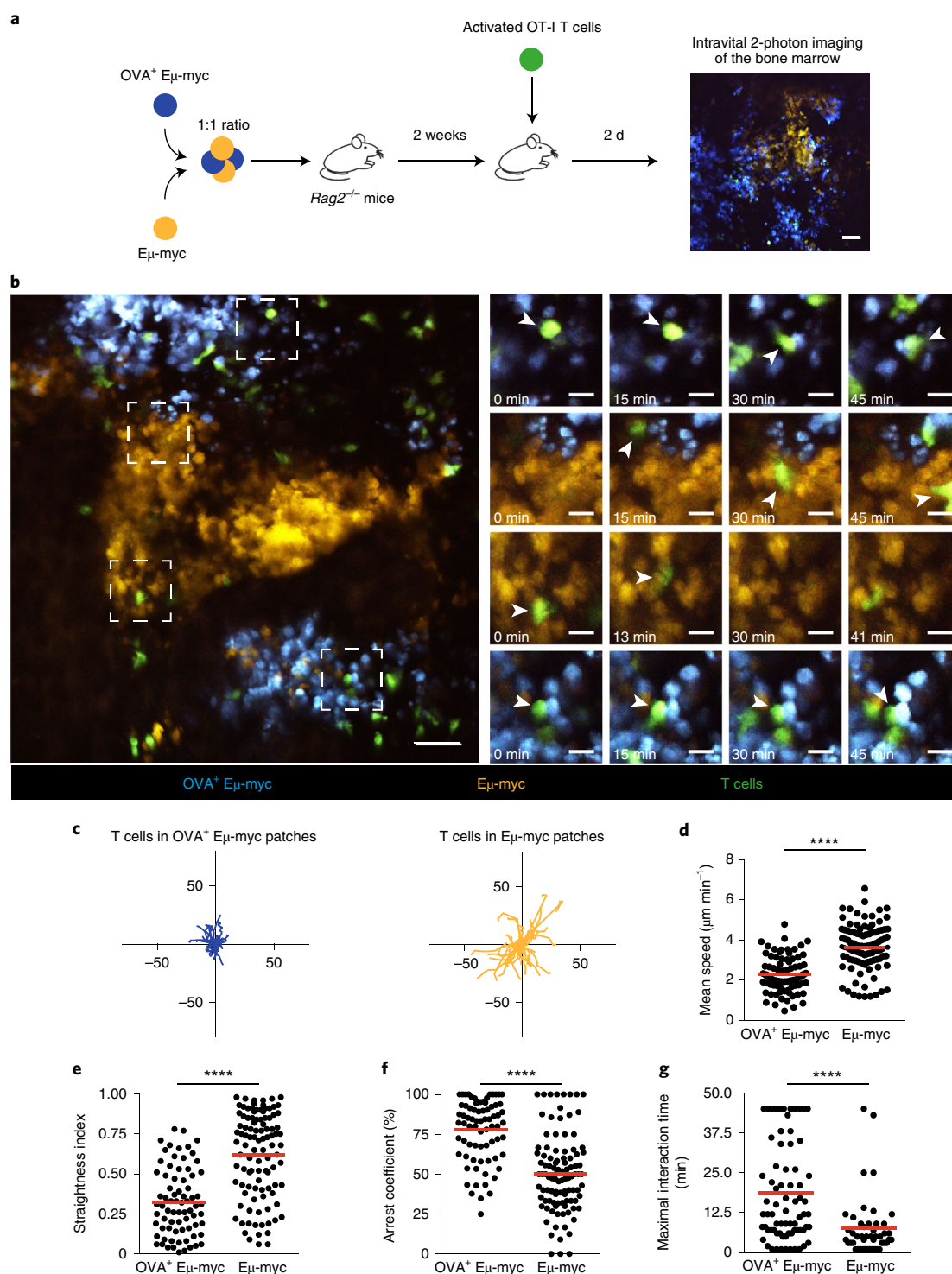


Fig. 2 | Tumor antigen expression drives the selective accumulation and arrest of intratumoral T cells. **a**, Experimental setup. *Rag2*^{-/-} mice were injected with a 1:1 mixture of OVA-expressing and OVA-non-expressing Eμ-myc B lymphoma cells, labeled with CFP and YFP, respectively. On days 12–13, mice were injected with activated GFP⁺ OT-I T cells. After 2 d, intravital imaging of the bone marrow was performed. Scale bar, 50 μm. **b–g**, CD8⁺ OT-I T cells specifically accumulate and arrest in antigen-expressing cellular patches of mosaic tumors. **b**, Left: representative image of OVA⁺ (blue) or OVA⁻ (orange) tumor patches infiltrated with OT-I T cells (green). Scale bar, 50 μm. Right: time-lapse images (corresponding to the dashed squares) showing OT-I T cells (indicated by arrows) forming stable contacts with OVA⁺ Eμ-myc cells but not with OVA⁻ Eμ-myc cells. Scale bars, 15 μm (representing two independent experiments with *n* = 3 mice in each experiment). **c**, Individual tracks of OT-I T cells located either in OVA⁺ tumor patches (blue) or OVA⁻ tumor patches (orange). Numbers indicate distance in μm. **d–f**, Graphs show mean speed (**d**), straightness (**e**) and arrest coefficient (**f**) of individual OT-I T cells located in OVA⁺ or OVA⁻ tumor patches. Only tracks that were at least 5-min long were considered. **g**, Maximal duration of interaction of individual OT-I T cells with OVA⁺ or OVA⁻ Eμ-myc cells measured during 45-min-long videos. Results in **c–g** are from five videos obtained in two independent experiments with three mice per experiment. In **d–g**, each dot represents one individual track (*n* = 75 OT-I tracks in OVA⁺ tumor patches; *n* = 103 OT-I tracks in OVA⁻ tumor patches). Red lines indicate mean values (*****P* < 0.0001, two-tailed Mann-Whitney *U*-test).

ing to a homogeneous response, the intensity of which depends on the number of producing T cells (Fig. 3e). To distinguish between these two possibilities, we repeated our experiments with different numbers of T cells. We reasoned that, if T cells acted on a limited number of tumor cells, lowering the number of T cells should reduce the frequency of responding tumor cells without affecting the intensity of their response (best modeled by a bimodal distribution in tumor cell response) (Fig. 3e). By contrast, if IFN- γ primarily acted as a shared resource, reducing the number of T cells should decrease the intensity of the response in all cells (unimodal distribution of tumor cell response). Supporting the latter hypothesis, we found that a tenfold decrease in the number of transferred T cells resulted in a homogeneous reduction in MHC class I upregulation (Fig. 3f). Altogether, these experiments support the idea that IFN- γ diffuses extensively in the tumor microenvironment. As a result, the collective activity of IFN- γ -producing T cells generates a widespread field of IFN- γ , the concentration of which dictates phenotypic changes across the whole tumor. We measured that, when transferring 20×10^6 T cells, the ratio of OVA⁺ E μ -myc cells to OT-I T cells was approximately 100:1. By comparing the fold increase in MHC class I levels observed in vivo with that seen in vitro (Fig. 1a), we estimated that this T cell density resulted in an intratumoral concentration of IFN- γ of around 1 ng ml^{-1} (see Extended data Fig. 5a,b). This is obviously a rough estimate because several parameters could potentially differ between the in vitro and the in vivo settings (such as cytokine consumption or basal IFN- γ receptor (IFN- γ R) expression). To test whether the activity of IFN- γ may influence tumor fate, we first assessed the effect of systemic IFN- γ delivery. We observed phenotypic changes in tumor cells, suggesting effective diffusion in vivo, and most importantly found reduced tumor burden in the bone marrow of treated mice (see Extended data Fig. 5c,d). This result suggests, that, in contrast to what was seen in vitro with E μ -myc cells, IFN- γ exerts a negative effect on tumor development in vivo. This discrepancy may originate from an indirect effect of IFN- γ on host cells or from changes in IFN- γ R expression/signaling on tumor cells in vivo. To test how T cell-derived IFN- γ may affect tumor growth in vivo, we relied on our mosaic tumor system and compared the impact of T cell-derived IFN- γ on the growth of both antigen-positive and antigen-negative tumors. When recipient mice were transferred with IFN- γ -competent T cells, we noted that antigen-positive (OVA⁺) cells were eliminated, but also that the antigen-negative tumor was partly controlled (see Extended data Fig. 5e,f). This partial control was lost on transfer of IFN- γ -deficient T cells (see Extended data Fig. 5e,f), supporting the idea that bystander IFN- γ activity may help limit tumor growth in vivo.

To extend these findings to solid tumors, we generated mosaic tumors using the B16.F10 melanoma model. In this setup, we again observed the hallmarks of IFN- γ signaling in both antigen-posi-

tive (OVA⁺ B16) and antigen-negative (B16) cells in the presence of T cells (Fig. 3g–i). Thus, extensive bystander activity of T cell-derived IFN- γ is a property that also pertains to solid tumors.

Widespread and sustained STAT1 signaling in the tumor microenvironment. The observed bystander changes in tumor phenotype prompted us to better characterize the spatiotemporal dynamics of IFN- γ signaling in the tumor microenvironment. We focused on STAT1 translocation in tumor cells, because this represents an early event after cytokine exposure. To this end, we generated a fluorescent reporter combining a STAT1–GFP (green fluorescent protein) fusion protein and a nuclear mCherry protein, and introduced this probe into E μ -myc lymphoma cells. As expected, in the absence of stimulation, STAT1–GFP was cytoplasmic and excluded from the nucleus (Fig. 4a). On in vitro addition of IFN- γ , a more uniform (cytoplasmic and nuclear localization) distribution of STAT1–GFP was rapidly observed, confirming the translocation of a fraction of the STAT1–GFP pool (Fig. 4a). To quantify STAT1 translocation in multiple cells, we designed an automated image analysis procedure based on the detection of cell and nuclear shape and computing of Pearson's correlation coefficients between STAT1–GFP and nuclear mCherry signals (see Extended data Fig. 6). With this strategy, statistically distinct translocation scores were observed for cells in the presence or absence of IFN- γ (Fig. 4b).

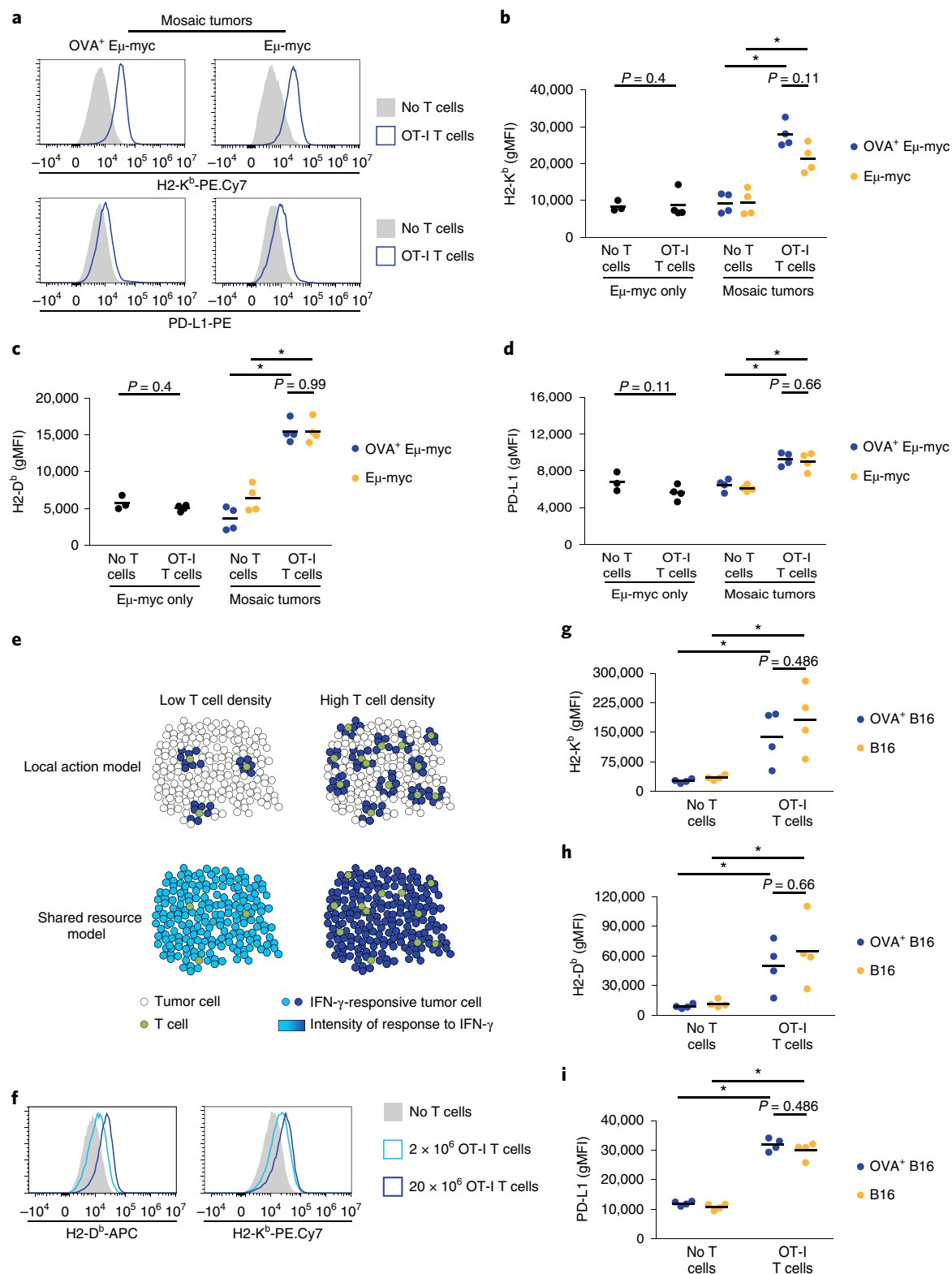
We therefore used these distinct patterns of STAT1 localization to identify cells with ongoing cytokine signaling in vivo by two-photon imaging. In a first model, we analyzed the response of H-Y⁺ B cell lymphoma in response to MataHari effector CD8⁺ T cells, which express a H-Y-specific transgenic TCR. In female *Rag2*^{−/−} mice with established tumor expressing the fluorescent reporter, we predominantly observed a cytoplasmic-only distribution of STAT1–GFP, suggesting an absence of IFN- γ signaling (Fig. 4c–e). However, in mice that were also transferred with tumor-reactive MataHari effector CD8⁺ T cells, we observed evidence of cytokine signaling in a large number of tumor cells as detected by STAT1 translocation. This was evident from two-photon images (Fig. 4c,d) and confirmed by automated computing of STAT1 translocation scores (Fig. 4e). Importantly, STAT1 translocation was widespread and detected in tumor cells irrespective of whether or not they were contacting T cells. Consistently, there was no correlation between STAT1 translocation score in individual tumor cells and distance to the nearest T cells (Fig. 4f). Nuclear STAT1–GFP typically persisted throughout the imaging experiments (typically 1–2 h), suggesting sustained signaling (Fig. 4g, and see Supplementary Video 2). To extend these results, we repeated these experiments in a second model relying on OVA⁺ E μ -myc B cell lymphoma and OT-I CD8⁺ T cells. In this setup also, nuclear STAT1–GFP was detected in the vast majority of tumor cells when OT-I effector CD8⁺ T cells were transferred but not in the absence of T cells (see Extended data

Fig. 3 | Extensive bystander IFN- γ activity in the tumor microenvironment. a–f, Bystander activity of IFN- γ in the tumor microenvironment of B cell lymphomas. *Rag2*^{−/−} mice were injected with either E μ -myc cells alone or a 1:1 mixture of E μ -myc and OVA-expressing E μ -myc B lymphoma cells (labeled with different fluorescent proteins). On days 12–13, recipients were injected with OT-I T cells or left untreated. After 2 d, the bone marrow of the mice was recovered and analyzed by flow cytometry. **a**, Representative histograms of H2-K^b and PD-L1 surface expression on OVA-expressing (left) or OVA-non-expressing (right) E μ -myc B lymphoma cells isolated from mice bearing mosaic tumors and injected with OT-I T cells (blue line) or left untreated (filled gray). Data represent three independent experiments with $n = 4$ mice per group in each experiment. **b–d**, H2-K^b (**b**), H2-D^b (**c**) and PD-L1 (**d**) surface expression on tumor cells from mice injected with E μ -myc cells only (black) or a 1:1 mixture of OVA-expressing (blue) and OVA-non-expressing (orange) E μ -myc cells. Each dot represents one mouse. Black lines indicate mean values. Data represent three independent experiments with $n = 4$ mice per group in each experiment (* $P < 0.05$, two-tailed Mann–Whitney *U*-test). **e**, Proposed models for IFN- γ diffusion in the tumor microenvironment (see text for details). **f**, Representative histograms of surface expression of H2-D^b and H2-K^b on OVA-expressing tumor cells isolated from mice either left untreated (filled gray) or injected with 2×10^6 (light-blue line) or 20×10^6 (dark-blue line) OT-I T cells. Data represent $n = 5$ mice per group. **g–i**, Bystander activity of IFN- γ in solid tumors. *Rag2*^{−/−} mice were injected with a 1:1 mixture of B16 and OVA-expressing B16 cells (labeled with different fluorescent proteins). After 1 week, recipients were injected with OT-I T cells or left untreated. After 2 d, tumors were digested and analyzed by flow cytometry. H2-K^b (**g**), H2-D^b (**h**) and PD-L1 (**i**) surface expression is shown on tumor cells from mice injected with a mixture of OVA-expressing (blue) and OVA-non-expressing (orange) B16.F10 cells. Each dot represents one mouse. Black lines indicate mean values. Data represent two independent experiments with $n = 4$ mice per group in each experiment (* $P < 0.05$, two-tailed Mann–Whitney *U*-test).

Fig. 7a–d and Supplementary Video 3). To further demonstrate the ability of T cell-derived IFN- γ to signal in distant tumor cells, we created mosaic tumors containing antigen-negative tumor that expressed the STAT1–GFP reporter and fluorescently labeled antigen-positive tumor cells. As shown in Extended data Fig. 7e–h, we detected increased STAT1 activity in antigen-negative tumor cells when T cells and antigen-positive tumor cells were present compared with controls lacking antigen-positive tumor cells.

These results establish that STAT1 activity occurs at the tissue level rather than in discrete spots and indicate that STAT1 translocation does not require ongoing interactions with effector T cells.

Sustained IFN- γ signaling is required for alteration of tumor cell phenotype. Early events of IFN- γ signaling, including STAT1 phosphorylation and STAT1 translocation, are typically detected within minutes of cytokine exposure³⁰. However, it is unclear whether



transient or prolonged exposure to IFN- γ is required to alter tumor cell phenotype. We therefore designed an experiment in which E μ -myc B lymphoma cells are cultured for a fixed period of time (24 h), but exposed to IFN- γ during this period for variable amounts of time (by washing out the cytokine and adding a blocking anti-IFN- γ monoclonal antibody to prevent any residual effects). RNA-sequencing (RNA-seq) was then performed for these different stimulation patterns (Fig. 5a). We confirmed that sustained IFN- γ exposure resulted in prolonged STAT1 translocation (see Extended data Fig. 8). As shown in Fig. 5b,c, the core of the response that included well-known IFN- γ -responsive genes (*Cd74*, *Cd274*, *H2-K1*, *H2-D1*, and so on) required at least 6–24 h of cytokine exposure. Notably, cells exposed to IFN- γ for 1 h were barely distinguishable from unstimulated cells. We confirmed these findings at the protein level for H2-K^b, H2-D^b and PD-L1 using flow cytometry (Fig. 5d). In these settings, 6–24 h of cytokine exposure was needed for substantial upregulation of these molecules. The need for long-lasting cytokine exposure was not restricted to B lymphoma cells because we observed a similar requirement for B16.F10 melanoma cells (Fig. 5e). Altogether, these results suggested that sustained exposure to IFN- γ is needed to drive phenotypic changes in tumor cells, including MHC class I and PD-L1 upregulation. Our transcriptomic analysis also revealed a strong enrichment of interferon regulatory factor 8 (IRF8) and STAT1 motifs and that the expression of both transcription factors, IRF8 and STAT1, increased in response to IFN- γ (Fig. 5f,g). This raises the possibility that a two-step signaling activation, in which STAT1 and IRF8 upregulation by IFN- γ subsequently potentiates the cellular response to this cytokine, accounts for the requirement for sustained cytokine exposure.

Probing IFN- γ activity in human melanoma. To test whether bystander IFN- γ activity also occurs in human tumors, we reanalyzed single-cell RNA-seq data from the immune infiltrate of eight non-treated melanoma patients³¹. Overall, we found that CD8⁺ T cells were the main producers of IFN- γ because more than 80% of all IFN- γ mRNA molecules were specific to those cells (Fig. 6a). We decided to investigate the impact of IFN- γ production on the tumor microenvironment by focusing on the monocyte/macrophage population as a sensor of intratumoral IFN- γ . Of note, the low number of tumor cells present in the samples precluded us from performing the same analysis on those cells. By performing over-dispersion analysis³², we identified one module (noted as pathway 7 in Extended data Fig. 9) specifically enriched in IFN- γ -responsive genes (Fig. 6b). Remarkably, the mean IFN- γ signature in monocytes/macrophages significantly correlated ($R=0.91$, $P=0.0018$; Fig. 6c) with the frequency of IFN- γ -producing CD8⁺ T cells.

Similarly, when repeating the analysis with intratumoral neutrophils, we observed an IFN- γ signature that correlated with the fraction of IFN- γ -producing T cells (Fig. 6e,f). These data suggest that, as observed in our mouse models, IFN- γ signals in multiple cell types of the tumor microenvironment. The observation that neutrophils, which are not typical tumor antigen-presenting cells, also harbor the IFN- γ signature suggests that they have received IFN- γ signals in a bystander fashion.

We also noted that, in patients with a high density of IFN- γ -producing T cells, the distribution of the IFN- γ signature on monocytes and neutrophils appeared unimodally shifted toward higher values (Fig. 6g). This observation is compatible with relatively uniform IFN- γ activity in the tumor, which was also observed in head and neck squamous cell carcinoma (HNSCC) patients, as analyzed by single-cell RNA-seq in tumor cells (see Extended data Fig. 10). Future studies assessing the spatial distribution of IFN- γ -responsive genes in relation to T cell infiltration will help refine our understanding of cytokine propagation and signaling in human tumors.

Discussion

In the present study, we established the spatiotemporal dynamics of intratumoral T cell-derived IFN- γ in models of B cell lymphoma and melanoma. We provided evidence that the collective activity of effector T cells, together with cytokine diffusion, drives a tumor-wide response. Moreover, the formation of a sustained field of IFN- γ was required to alter tumor cell phenotype.

The range of cytokine activity is dependent on the production rate and levels, diffusion and consumption by other cells¹⁹. Our results support the idea that T cell production of IFN- γ generates a widespread cytokine field shared by most tumor cells and tumor-infiltrating immune cells. MHC class I expression was uniformly upregulated in the tumor microenvironment even in cells located at a distance (>100 μ m) from the sites of cognate T cell interactions, a feature that may favor tumor lysis by cytotoxic T cells. Conversely, the expression of the inhibitory ligand PD-L1 was also upregulated by IFN- γ in a bystander fashion. Thus, both potentially beneficial and deleterious effects of IFN- γ appeared to be regulated by a cytokine field, rather than by very discrete cytokine hotspots.

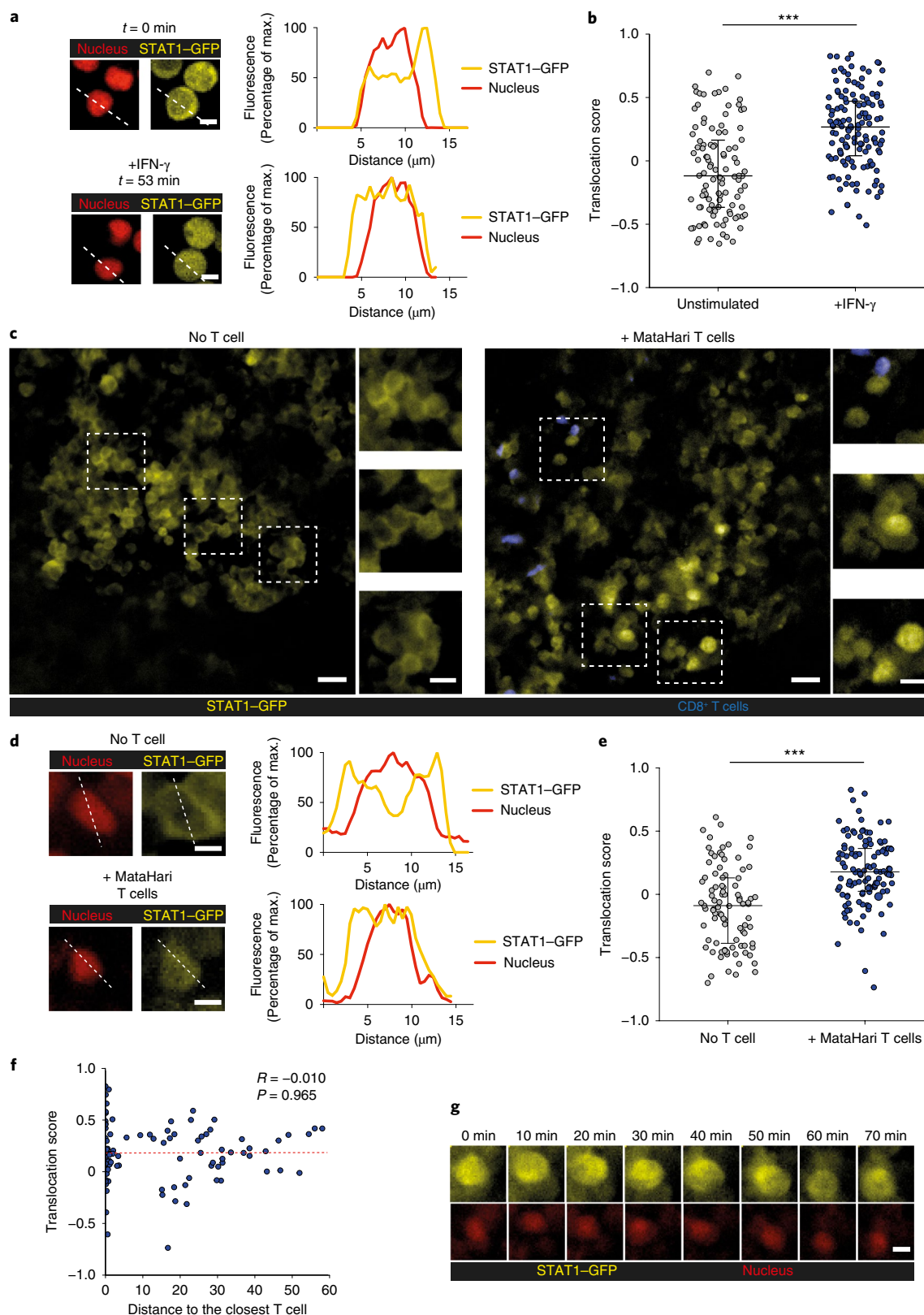
At first, these results may appear at odds with the well-established notion that IFN- γ is produced by T cells at the immunologic synapse with target cells^{20,33,34}, a phenomenon that should generate high local concentrations of IFN- γ at sites of antigen recognition and create extensive heterogeneity in cell responses. Our finding that prolonged exposure to IFN- γ is required to alter tumor cell phenotype probably provides an explanation for this apparent contradiction because these cytokine hotspots will probably dissipate

Fig. 4 | T cells mediate widespread and sustained STAT1 activity in the tumor microenvironment. **a,b**, Male E μ -myc B lymphoma cells were transduced to express a STAT1-GFP reporter and a nuclear mCherry protein. STAT1 activity is measured on E μ -myc B cells cultured in the absence or presence of recombinant IFN- γ . **a**, Images and graphs depict the quantification of STAT1-GFP (yellow) and nuclear mCherry (red) fluorescence intensity across the indicated lines for a representative cell before (top) and after (bottom) IFN- γ exposure. Scale bar, 5 μ m. **b**, Translocation score was computed automatically for cells before or after IFN- γ exposure. Each dot represents one cell (unstimulated $n=106$ cells; IFN- γ $n=135$ cells; *** $P<0.001$, two-sided Tukey's range test; error bars indicate mean $\pm$ s.d.). **c–g**, Recipient female *Rag2*^{-/-} mice were injected with H-Y⁺ E μ -myc B lymphoma cells expressing the STAT1-GFP reporter and a nuclear mCherry protein. After 3 weeks, activated CD8⁺ T cells bearing the anti-H-Y MataHari TCR were injected i.v. After 3 d, recipients were subjected to intravital imaging of the bone marrow. **c–e**, Detection of nuclear STAT1-GFP in T cell-infiltrated tumors. **c**, Representative two-photon images (scale bar, 20 μ m), highlighting three specific regions (insets, scale bar, 10 μ m) of the tumor in mice either transferred or not transferred with MataHari T cells. **d**, Quantification of STAT1-GFP (yellow) and nuclear mCherry (red) fluorescence intensity across the indicated line for a representative tumor cell in mice either left untreated (top) or transferred with MataHari T cells (bottom). Scale bar, 5 μ m. **e**, The translocation score was computed automatically from two-photon images obtained in mice left untreated (no T cells) or transferred with MataHari T cells. Each dot represents one tumor cell (no T cell $n=92$ cells; MataHari T cells $n=115$ cells; *** $P<0.001$, two-sided Tukey's range test; error bars indicate mean $\pm$ s.d.). **f**, STAT1 translocation in tumor cells is largely independent of the distance to the nearest T cells. The translocation score is graphed for each tumor cell as a function of the calculated distance to the nearest T cell. Each dot represents one cell ($n=115$ cells; R represents Pearson's correlation coefficient; statistical significance was assessed using Fisher's test). **g**, STAT1-GFP translocation is sustained in vivo. Representative time-lapse images show the maintenance of STAT1-GFP translocation during the imaging period. Scale bar, 5 μ m. Data shown in **b–g** represent two independent experiments with three mice per group in each experiment.

due to diffusion well before they have the time to generate a local biological effect.

A direct consequence of this mode of action is that the activity of IFN- γ will be determined at the tissue level by the overall number of cytokine-producing cells. In this model, the collective

activity of T cells is essential because it determines the overall cytokine concentration and the extent of tumor cell response, in a manner resembling quorum sensing. Such a collective mode of action was recently described for nitric oxide (NO)-producing macrophages, where a single macrophage has a negligible intrinsic



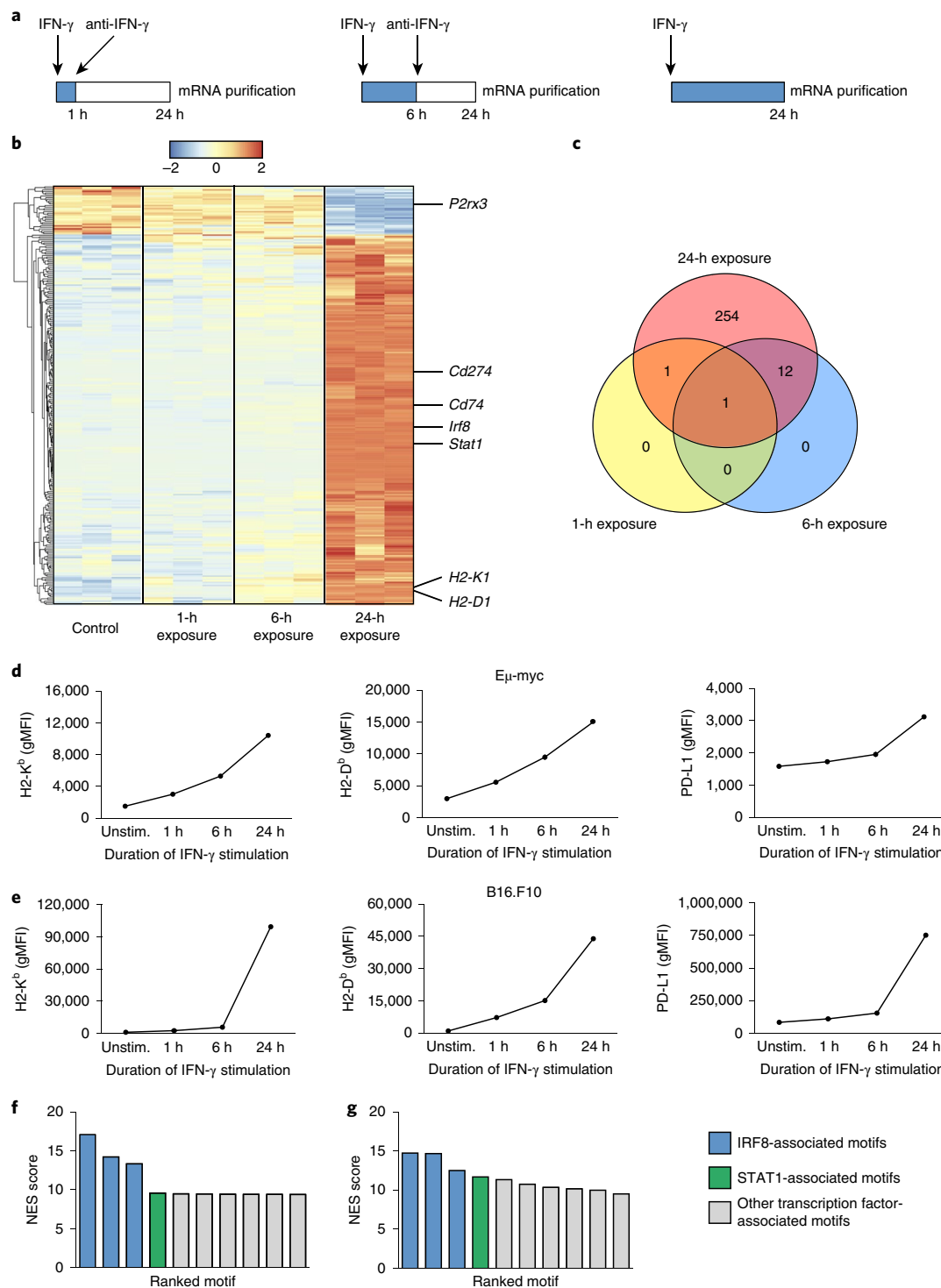


Fig. 5 | Sustained signaling is required to alter tumor cell phenotype. a, Experimental setup. Eμ-myc B lymphoma cells were stimulated with 5 ng ml⁻¹ of IFN-γ. At 1 h or 6 h, the stimulation was blocked by adding 50 μg ml⁻¹ of anti-IFN-γ monoclonal antibody. At 24 h, cells were recovered for mRNA-seq. **b**, Heatmap of differentially expressed genes. Gene expression is normalized by row. **c**, Venn diagrams of differentially expressed genes between the various stimulated and control samples. **d,e**, Tumor cells were stimulated with 5 ng ml⁻¹ of IFN-γ. At 1 h or 6 h, the stimulation was blocked by adding 50 μg ml⁻¹ of anti-IFN-γ monoclonal antibody. At 24 h, cells were recovered and analyzed by flow cytometry. Graphs represent H2-K^b (left), H2-D^b (middle) and PD-L1 (right) surface expression after indicated durations of stimulation on Eμ-myc (**d**) and B16.F10 (**e**) models. Each dot represents the mean of three technical replicates, representing three independent experiments. Unstim., unstimulated. **f,g**, Normalized enrichment score (NES) of the 10 most enriched motifs for the differentially expressed genes between 6-h-stimulated (**f**) and 24-h-stimulated (**g**) samples and control samples.

activity, but a high number of NO-producing macrophages regulate activity at the tissue level in both NO-producing and NO-non-producing cells^{35,36}.

It has recently been shown that IFN-γ can bind to phosphatidylserine on the surface of tumor cells and be slowly released to prolong inflammation³⁰. Such a mechanism, termed ‘catch and release’

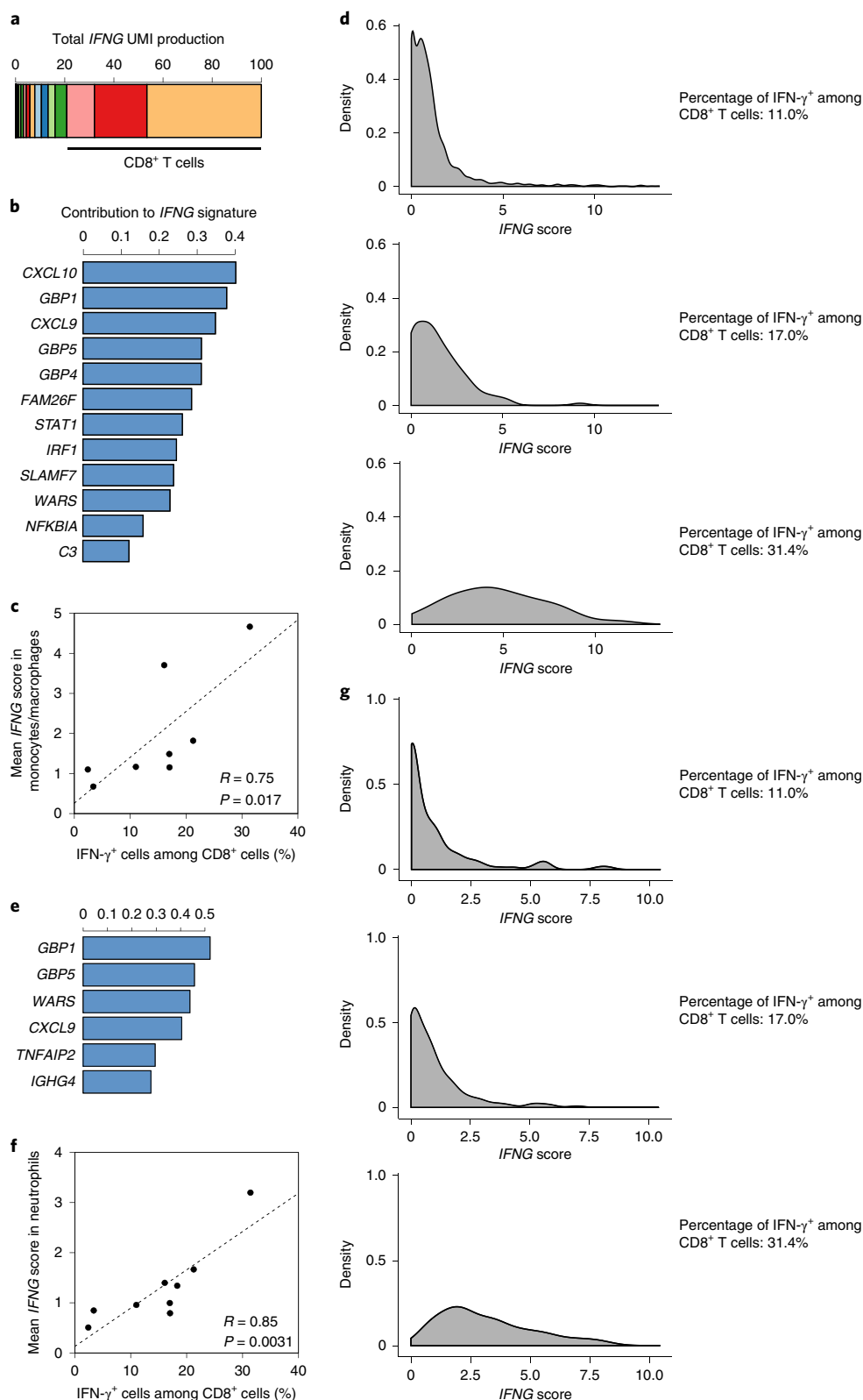


Fig. 6 | Assessing IFN- γ bystander activity in human melanoma samples. Measurement of the IFN- γ signature in the tumor microenvironment of melanoma patients using single-cell RNA-seq. **a**, Contribution of distinct cell clusters to *IFNG* RNA unique molecular identifier (UMI) production. Data are compiled from all patients analyzed ($n=8$). **b**, Gene contribution to the IFN- γ signature in the monocyte/macrophage cluster. **c**, Association between the percentage of IFN- γ cells (>1 UMI) among CD8⁺ cells and the IFN- γ score in monocytes/macrophages (R represents Pearson's correlation coefficient; statistical significance was assessed using Fisher's test). **d**, Distribution of the IFN- γ signature in monocytes/macrophages from three different patients harboring distinct indicated frequencies of IFN- γ CD8⁺ T cells (density in a.u., arbitrary units). **e**, Gene contribution to the IFN- γ signature in the neutrophil cluster. **f**, Association between the percentage of IFN- γ cells among CD8⁺ cells and the IFN- γ score in neutrophils (R represents Pearson's correlation coefficient; statistical significance was assessed using Fisher's test). **g**, Distribution of IFN- γ signature in neutrophils from three different patients harboring distinct indicated frequencies of IFN- γ CD8⁺ T cells.

may favor the formation of relatively stable cytokine fields by temporarily averaging cytokine availability in the tumor microenvironment. Binding of IFN- γ on the extracellular matrix should also achieve similar effects. It will be important to determine whether other cytokines rely on a similar global mode of signaling and assess how the superimposition of distinct cytokine fields may translate into complex biological responses at the tissue level, in particular in the context of the tumor microenvironment. This is particularly relevant for TNF- α which can act synergistically with IFN- γ to induce tumor cell death. Cytokine consumption plays an important role in shaping the spatial range of cytokine activity³⁷, and may generate fields of different sizes for distinct cytokines, possibly creating functional heterogeneity in the tissue.

In summary, our work provides new insights into the spatiotemporal activity of intratumoral IFN- γ with important implications for anti-tumor immune responses. Single-cell RNA-seq, as illustrated here, but also spatial transcriptomics^{38–40}, will help estimate the range of cytokine activity in human tumors. Finally, the use of systems immunology, together with the development of new probes to image cytokine propagation and activity in vivo, will be essential to establish the rules of cytokine-mediated communication at steady-state or during disease pathogenesis.

Methods

Mice and cell lines. The 6- to 8-week-old C57BL/6 mice were purchased from Charles River. *Rag2*^{-/-}, *Rag2*^{-/-} γ_c ^{-/-}, *Rag1*^{-/-} OT-I TCR, UBC-GFP *Rag1*^{-/-} OT-I TCR, cyan fluorescent protein (CFP expressed under the actin promoter), *Rag2*^{-/-} MataHari TCR and *Ifng*^{-/-} mice were bred and crossed in our animal facility. All experiments were performed in agreement with relevant guidelines and regulations and approved by the Institut Pasteur Committee on Animal Welfare (CETEA) under protocol code 170038. A lymphoma cell line was isolated from tumor-bearing, male, E μ -myc transgenic mice²⁶, which develop spontaneous Burkitt-like lymphomas. This cell line was retrovirally transduced to express CFP, GFP, yellow fluorescent protein (YFP), membrane CFP, mCherry-NLS fluorescent proteins, ovalbumin or STAT1–GFP fusion protein (all cloned in an MSCV backbone). B16.F10 (B16) melanoma cells were kindly provided by G. Shakhar.

Flow cytometry and antibodies. For ex vivo analyses, femurs and tibias were isolated from E μ -myc tumor-bearing mice. Bone marrow cells were extracted by flushing the bones with phosphate-buffered saline supplemented with 1 mM ethylenediaminetetraacetic acid and 0.5% fetal bovine serum (FACS buffer), and single-cell suspensions were prepared by filtering the cells through 70- μ m cell strainers. For B16 tumor-bearing mice, tumors were harvested and cells were recovered by digesting the tumors with collagenase/DNase for 30 min at 37°C, before crushing them on 70- μ m cell strainers. All cells were then stained in FACS buffer, supplemented with 4% normal mouse and rat sera and/or 10 μ g ml⁻¹ of anti-mouse CD16/32 (Fc block), using a combination of fluorescent or biotin-labeled monoclonal antibodies from the following: CD8 (53-6.7, BD Biosciences), CD11b (BioLegend, M1/70), CD45 (BD Biosciences, 30-F11), H2-K^b (BioLegend, AF6-88.5), H2-D^b (BioLegend, KH95), Ly6C (BioLegend, HK1.4), Ly6G (BioLegend, 1A8), NK1.1 (BioLegend, PK136) and PD-L1 (BioLegend or BD Biosciences, 10F9G2 or MIH5, respectively). IFN- γ intracellular staining was performed using the Cytofix/Cytoperm kit (BD Biosciences) according to the manufacturer's guidelines and XMG1.2 monoclonal antibody (eBiosciences). IFN- γ expression was assayed in the presence of 1 μ g ml⁻¹ of Brefeldin A. Samples were analyzed using a BD FACSCanto II or a Cytoflex LX (Beckman Coulter) flow cytometer. Data analysis was performed using FlowJo v.10.4.1 (TreeStar).

Tumor models and T cell adoptive transfer. B cell lymphomas were established by injecting 1 $\times 10^6$ E μ -myc cells intravenously (i.v.) into the indicated recipients. Melanoma tumors were established by inoculating 2 $\times 10^6$ B16 cells subcutaneously. Splenocytes from MataHari TCR transgenic mice or OT-I TCR transgenic mice were isolated and red blood cells were removed by ammonium–chloride–potassium lysis. One-third of the cells was then pulsed with 50 μ M H-2D^b-restricted Uty_{246–254} peptide (WMHHNMLDI) or H-2K^b-restricted OVA_{257–264} peptide (SIINFELK) for 2 h in RPMI medium 1640–Glutamax at 37°C. The rest of the cells were incubated at 37°C in complete medium supplemented with 50 μ g ml⁻¹ of gentamicin. The two populations were mixed and cultured for 2–3 d. Cells were then subjected to Ficoll gradient centrifugation, to remove dead cells, and cultured in complete medium containing human IL-2 (25 IU ml⁻¹; Roche or R&D). For proper in vitro activation, IFN- γ -deficient T cells were supplemented with 100 ng ml⁻¹ of IFN- γ (Invitrogen). Cells were used on days 4–8, at which time >95% were CD8⁺ T cells. When indicated, T cells were retrovirally transduced to express mCherry as described⁴¹. Tumor-bearing mice were adoptively transferred

with cultured T cells by intravenous injection of 20 $\times 10^6$ cells. For experiments using CAR T cells, B cell lymphoma injection, conditioning and CAR T cell transfer were performed as previously described⁴¹.

Intravital two-photon imaging. Bone marrow imaging was performed as described⁴¹. Mice were anaesthetized with a mixture of xylazine (Rompun, 10 mg kg⁻¹) and ketamine (Imalgène, 100 mg kg⁻¹), which was replenished hourly. During imaging, the mice temperature was maintained at 37°C with a heating pad. Two-photon imaging was performed with an upright microscope FVMPE-RS (Olympus) and a $\times 25/1.05$ numerical aperture, water-dipping objective (Olympus). Excitation was provided by an Insight DeepSee dual laser (Spectra-Physics) tuned at 920 or 960 nm. The following filters were used for fluorescence detection: CFP (483/32), GFP (520/35), YFP (542/27), background (593/35) and mCherry (624/40). To create time-lapse sequences, we typically scanned a 30- to 40- μ m-thick volume of tissue at 5- μ m Z-steps and 60-s intervals.

Image analysis. Videos were processed and analyzed using Imaris software (Bitplane) or Fiji software (ImageJ 1.50 i). Videos and figures based on two-photon microscopy are shown as two-dimensional maximum intensity projections of three-dimensional data. For optimal contrast rendering, GFP was pseudo-colored in yellow in some images. Contact durations were calculated manually, based on close apposition between a given T cell and a tumor cell. To perform an unbiased analysis of STAT1–GFP translocation, we developed an in-house MATLAB script that first automatically detects cell envelope and cell nuclei, and then computes a translocation score. Briefly, both STAT1–GFP and nuclear mCherry channels are first binarized using Otsu's method. Image closing with a circular structuring element is then applied to the STAT1–GFP binarized image, followed by a filling of the holes. Euclidean distance transform is then applied to both binarized images before using a watershed segmentation tool to segment the contiguous cells. For each identified cell, Pearson's correlation between the STAT1–GFP and the mCherry signal is computed and considered as a good proxy for STAT1–GFP translocation (translocation score). This strategy was applied to both in vitro and in vivo data; however, due to the increased background signal in in vivo images, we added a step of contrast enhancement before image binarization: first, a Laplacian of Gaussian filter with a sigma parameter of size equal to the estimated size of the cells is used, followed by the extraction of the local peak with an *h*-dome transform⁴². Whenever the automated segmentation was not sufficiently accurate, we performed manual cell segmentation and computed Pearson's correlation coefficient using the region of interest manager and coloc2 Fiji plugins.

Messenger RNA-seq and analysis. The mRNA-seq was performed on cultured E μ -myc cells. The mRNAs were extracted using the RNeasy Mini-Kit (Qiagen) according to the manufacturer's guidelines. RNA-seq libraries were prepared using the TruSeq Stranded mRNA Library Preparation Kit, in accordance with the manufacturer's instructions. Briefly, 500 ng of total RNA was used for purification and fragmentation of mRNA. Purified mRNA underwent first- and second-strand complementary DNA synthesis. Complementary DNA was then adenylated, ligated to Illumina sequencing adapters and amplified by PCR (using 10 cycles). Final libraries were evaluated using fluorescent-based assays including PicoGreen (Life Technologies) or Qubit Fluorometer (Invitrogen) and Fragment Analyzer (Advanced Analytics) or BioAnalyzer (Agilent 2100), and were sequenced on an Illumina HiSeq2500 sequencer (v.4 chemistry) using 2 $\times$ 125 basepair (bp) cycles.

Ribosomal RNA abundance was measured by mapping with Bowtie2. Quality control was performed using Picard and RSeQC. The reads were aligned with STAR (v.2.5.2a), and genes annotated in Gencode v.M16 were quantified with featureCounts (v.1.4.3-p1). Normalization and differential expression were done with the Bioconductor package DESeq2 (v.1.18.1). Differential expression analysis, as well as normalization, were performed using the DESeq2 package (v.1.18.1) on R (v.3.4.3). Analysis was performed according to the recommendation of the package with the same parameter, except for the log₂(fold change) shrinkage method parameter that was set to 'apeglm'. A gene was considered to be significantly differentially expressed if its corresponding adjusted *P* value was <0.01 and its absolute shrinked log₂(fold change) > 1. Expression heatmap was performed using the pheatmap package, with Ward's criterion used for the hierarchical clustering.

Enrichment of specific transcription factor-binding motifs in the promoter of upregulated genes was performed using the iRegulon Cytoscape plugin⁴³. *Mus musculus* was chosen as the reference species and all other parameters were set to default, except the putative regulatory region and motif-ranking databases that were set to '500-bp upstream'.

Single-cell RNA-seq data clustering and analysis. The two single-cell RNA-seq datasets were downloaded on Gene Expression Omnibus (GEO) repository (accession numbers GSE123139 and GSE103322 for the melanoma and HNSCC data, respectively). Annotations were loaded using the GEOquery package (v.2.48.0). As the HNSCC data were already processed (provided as log(TPM + 1) data; TPM, transcripts per million), we used two slightly different settings for data pre-processing. In the case of the melanoma data, we processed as follows: to avoid removing populations of small-sized cells, a threshold of relatively low number of unique molecules (350 unique molecular identifiers or UMIs) was used in the first

instance to filter out low-quality cells. Genes with fewer than 300 UMIs expressed were also removed to reduce computation cost and time. The filtered data were then analyzed using the PAGODA2 pipeline (<https://github.com/hms-dbmi/pagoda2>)⁴⁴. Expression values and variance were scaled as described in Lake et al.⁴⁴. It is important to note that, although the first version of PAGODA relies on gene set analysis to perform cell clustering, PAGODA2 performs most of the analytical steps without any gene set analysis. The general approach used is similar to the one described at <https://github.com/hms-bmi/pagoda2/blob/master/vignettes/pagoda2.walkthrough.oct2018.md>. First gene variance was corrected and highly variable genes identified using the `adjustVariance` function (`gam.k=10`). The 100 first components of principal component analysis (PCA), based on the 3,000 most variable genes, were then computed using the `calculatePcaReduction` function. Then a *K*-nearest-neighbors (KNN) graph was computed based on the PCA reduction through the `makeKnnGraph` function (`k=30`). Cosine distance was used at this stage to increase the robustness of the analysis. Last, Louvain community detection clustering was performed on the KNN graph using the `getKnnClusters` function and the `multilevel.community` functions from the `igraph` R package. Once the cells were clustered, we identified all genes that were significantly expressed by cells from a given cluster compared with all other cells identified using the PAGODA2 function `getDifferentialGenes`, which performs the Kruskal–Wallis test, with default parameters (`corrected Z-score > 3`, `upregulated.only=TRUE`). Using this gene list, we annotated the various cell clusters based on the specific expression of well-known immune markers (CD3, CD8A, TRBC2, and so on). Low-dimensional embedding of the data was performed using the UMAP algorithm⁴⁵. The `uwot` package implementation was used (<https://github.com/jlmelville/uwot>). PCA dimensionality reductions computed using PAGODA2 were used as input data. The `N_neighbors` parameter was set to 40 and spread parameter to 4, and a cosine metric was used. In the case of the HNSCC, the data were already normalized and loaded into a `Pagoda2` object with parameter `'modelType'` set to `'raw'`. For the next steps of the analysis, the 50 first PCs were used, the 1,000 most variable genes were selected, the *K* parameter was set to 15 and a cosine distance was used.

Analysis of over-dispersed pathways. To detect biologically meaningful gene modules/pathways in a robust manner, we performed pathway over-dispersion analysis using *de novo* gene sets. We first selected a given cell population and performed gene expression and variance normalization on the raw UMI counts as described in the previous section. The variable genes were selected using the `getDifferentialGenes` function from the PAGODA2 package with the 200 top genes selected. These genes were then grouped into 15 pathways using hierarchical clustering (Ward linkage and correlation-based distance). Over-dispersion of these pathways was computed using the `testPathwayOverdispersion` function from the same package with default parameters. Gene set enrichment was performed through a binomial test (`binom.test` function, `alternative = 'greater'`) using the 200 most variable genes as the reference 'universe'. The 'HALLMARK INTERFERON ALPHA RESPONSE' and 'HALLMARK INTERFERON GAMMA RESPONSE' gene sets were downloaded from the Gene Set Enrichment Analysis website (<https://www.gsea-msigdb.org/gsea/msigdb/genesets.jsp?collection=H>). Multiple testing correction was performed using the Benjamini–Hochberg method (`p.adjust` R function).

Association between IFN- γ score and IFN- γ production. In the case of the melanoma dataset, the macrophage population was identified as the C1Q/APOE-expressing cluster. The IFN- γ score was computed using the `testPathwayOverdispersion` function as described for the macrophage population and its mean value computed for each patient. Neutrophils were identified as the S100A8/S100A9/Lyz-expressing cluster. Patients with fewer than 25 macrophages or neutrophils were removed from the analysis. The fraction of IFN- γ -expressing CD8⁺ T cells among T cells was computed by computing the number of cells annotated as coming from a 'CD3' gating belonging to a CD8⁺ T cell RNA-seq cluster, and has more than 0 UMI coming from the IFN- γ gene. The association between the two was assessed using a simple linear regression model (`lm` R function). The significance of the association was checked using Fisher's test (analysis of variance R function). Correlation was estimated using the `cor` R function. For HNSCC samples, patients with fewer than 50 tumor cells in the dataset were removed from the analysis.

Statistics and reproducibility. All statistical tests were performed using GraphPad Prism v.6. Points in graphs indicate individual mice, and lines indicate means. In bar graphs, bars indicate means and error bars indicate sample s.d. The statistical tests employed are detailed in the figure legends. No statistical method was used to predetermine sample size. No data were excluded from the analyses. The experiments were not randomized. The investigators were not blinded to allocation during the experiments and outcome assessment.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

RNA-seq data reported in this paper are deposited in the National Center for Biotechnology Information GEO database (GSE140191). Previously published human

single-cell RNA-seq data that were reanalyzed in the present study are available under accession codes GSE123139 and GSE103322. All other data supporting the findings of the present study are available from the corresponding author on reasonable request.

Code availability

The R and Matlab scripts used for the transcriptomic and image analysis are available on GitHub (https://github.com/PierreBSC/IFNG_Cancer_project).

Online content

Any methods, additional references, Nature Research reporting summaries, source data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s43018-020-0038-2>.

Received: 28 May 2019; Accepted: 29 January 2020;

Published online: 09 March 2020

References

- Kaplan, D. H. et al. Demonstration of an interferon gamma-dependent tumor surveillance system in immunocompetent mice. *Proc. Natl Acad. Sci. USA* **95**, 7556–7561 (1998).
- Street, S. E., Cretney, E. & Smyth, M. J. Perforin and interferon-gamma activities independently control tumor initiation, growth, and metastasis. *Blood* **97**, 192–197 (2001).
- Dunn, G. P., Koebel, C. M. & Schreiber, R. D. Interferons, immunity and cancer immunoediting. *Nat. Rev. Immunol.* **6**, 836–848 (2006).
- Garris, C. S. et al. Successful anti-PD-1 cancer immunotherapy requires T cell–dendritic cell crosstalk involving the cytokines IFN-gamma and IL-12. *Immunity* **49**, 1148–1161 e1147 (2018).
- Gao, J. et al. Loss of IFN-gamma pathway genes in tumor cells as a mechanism of resistance to anti-CTLA-4 therapy. *Cell* **167**, 397–404 e399 (2016).
- Chin, Y. E. et al. Cell growth arrest and induction of cyclin-dependent kinase inhibitor p21 WAF1/CIP1 mediated by STAT1. *Science* **272**, 719–722 (1996).
- Ikeda, H., Old, L. J. & Schreiber, R. D. The roles of IFN gamma in protection against tumor development and cancer immunoediting. *Cytokine Growth Factor Rev.* **13**, 95–109 (2002).
- Braumüller, H. et al. T-helper-1-cell cytokines drive cancer into senescence. *Nature* **494**, 361–365 (2013).
- Wang, W. et al. CD8⁺ T cells regulate tumour ferroptosis during cancer immunotherapy. *Nature* **569**, 270–274 (2019).
- Kammertoens, T. et al. Tumour ischaemia by interferon-gamma resembles physiological blood vessel regression. *Nature* **545**, 98–102 (2017).
- Meunier, M. C. et al. T cells targeted against a single minor histocompatibility antigen can cure solid tumors. *Nat. Med.* **11**, 1222–1229 (2005).
- Groom, J. R. & Luster, A. D. CXCR3 ligands: redundant, collaborative and antagonistic functions. *Immunol. Cell Biol.* **89**, 207–215 (2011).
- Takeda, K. et al. IFN-gamma is required for cytotoxic T cell-dependent cancer genome immunoediting. *Nat. Commun.* **8**, 14607 (2017).
- García-Díaz, A. et al. Interferon receptor signaling pathways regulating PD-L1 and PD-L2 expression. *Cell Rep.* **19**, 1189–1201 (2017).
- Dong, H. et al. Tumor-associated B7-H1 promotes T-cell apoptosis: a potential mechanism of immune evasion. *Nat. Med.* **8**, 793–800 (2002).
- Freeman, G. J. et al. Engagement of the PD-1 immunoinhibitory receptor by a novel B7 family member leads to negative regulation of lymphocyte activation. *J. Exp. Med.* **192**, 1027–1034 (2000).
- Pai, C. S. et al. Clonal deletion of tumor-specific T cells by interferon-gamma confers therapeutic resistance to combination immune checkpoint blockade. *Immunity* **50**, 477–492 e478 (2019).
- Benci, J. L. et al. Tumor interferon signaling regulates a multigenic resistance program to immune checkpoint blockade. *Cell* **167**, 1540–1554 e1512 (2016).
- Altan-Bonnet, G. & Mukherjee, R. Cytokine-mediated communication: a quantitative appraisal of immune complexity. *Nat. Rev. Immunol.* **19**, 205–217 (2019).
- Huse, M., Lillemeyer, B. F., Kuhns, M. S., Chen, D. S. & Davis, M. M. T cells use two directionally distinct pathways for cytokine secretion. *Nat. Immunol.* **7**, 247–255 (2006).
- Helmstetter, C. et al. Individual T helper cells have a quantitative cytokine memory. *Immunity* **42**, 108–122 (2015).
- Honda, T. et al. Tuning of antigen sensitivity by T cell receptor-dependent negative feedback controls T cell effector function in inflamed tissues. *Immunity* **40**, 235–247 (2014).
- Sanderson, N. S. et al. Cytotoxic immunological synapses do not restrict the action of interferon-gamma to antigenic target cells. *Proc. Natl Acad. Sci. USA* **109**, 7835–7840 (2012).

24. Perona-Wright, G., Mohrs, K. & Mohrs, M. Sustained signaling by canonical helper T cell cytokines throughout the reactive lymph node. *Nat. Immunol.* **11**, 520–526 (2010).
25. Müller, A. J. et al. CD4⁺ T cells rely on bystander effector activity to control local infection. *Immunity* **37**, 147–157 (2012).
26. Harris, A. W. et al. The E mu-myc transgenic mouse. A model for high-incidence spontaneous lymphoma and leukemia of early B cells. *J. Exp. Med.* **167**, 353–371 (1988).
27. Hart, I. R. The selection and characterization of an invasive variant of the B16 melanoma. *Am. J. Pathol.* **97**, 587–600 (1979).
28. Sugiura, K. & Stock, C. C. Studies in a tumor spectrum. I. Comparison of the action of methylbis(2-chloroethyl)amine and 3-bis(2-chloroethyl)aminomethyl-4-methoxymethyl-5-hydroxy-6-methylpyridine on the growth of a variety of mouse and rat tumors. *Cancer* **5**, 382–402 (1952).
29. Milo, I. et al. The immune system profoundly restricts intratumor genetic heterogeneity. *Sci. Immunol.* **3**, eaat1435 (2018).
30. Oyler-Yaniv, J. et al. Catch and release of cytokines mediated by tumor phosphatidylserine converts transient exposure into long-lived inflammation. *Mol. Cell* **66**, 635–647 e637 (2017).
31. Li, H. et al. Dysfunctional CD8 T cells form a proliferative, dynamically regulated compartment within human melanoma. *Cell* **176**, 775–789 e718 (2019).
32. Fan, J. et al. Characterizing transcriptional heterogeneity through pathway and gene set overdispersion analysis. *Nat. Methods* **13**, 241–244 (2016).
33. Kupfer, A., Mosmann, T. R. & Kupfer, H. Polarized expression of cytokines in cell conjugates of helper T cells and splenic B cells. *Proc. Natl Acad. Sci. USA* **88**, 775–779 (1991).
34. Egen, J. G. et al. Intravital imaging reveals limited antigen presentation and T cell effector function in mycobacterial granulomas. *Immunity* **34**, 807–819 (2011).
35. Olekhovitch, R., Ryffel, B., Muller, A. J. & Bousso, P. Collective nitric oxide production provides tissue-wide immunity during leishmania infection. *J. Clin. Invest.* **124**, 1711–1722 (2014).
36. Postat, J., Olekhovitch, R., Lemaître, F. & Bousso, P. A metabolism-based quorum sensing mechanism contributes to termination of inflammatory responses. *Immunity* **49**, 654–665 (2018).
37. Oyler-Yaniv, A. et al. A tunable diffusion–consumption mechanism of cytokine propagation enables plasticity in cell-to-cell communication in the immune system. *Immunity* **46**, 609–620 (2017).
38. Stahl, P. L. et al. Visualization and analysis of gene expression in tissue sections by spatial transcriptomics. *Science* **353**, 78–82 (2016).
39. Shah, S., Lubeck, E., Zhou, W. & Cai, L. seqFISH accurately detects transcripts in single cells and reveals robust spatial organization in the hippocampus. *Neuron* **94**, 752–758 e751 (2017).
40. Wang, X. et al. Three-dimensional intact-tissue sequencing of single-cell transcriptional states. *Science* **361**, pii: eaat5691 (2018).
41. Cazaux, M. et al. Single-cell imaging of CAR T cell activity in vivo reveals extensive functional and anatomical heterogeneity. *J. Exp. Med.* **216**, 1038–1049 (2019).
42. Vincent, L. Morphological grayscale reconstruction in image analysis: applications and efficient algorithms. *IEEE Trans. Image Process.* **2**, 176–201 (1993).
43. Janky, R. et al. iRegulon: from a gene list to a gene regulatory network using large motif and track collections. *PLoS Comput. Biol.* **10**, e1003731 (2014).
44. Lake, B. B. et al. Integrative single-cell analysis of transcriptional and epigenetic states in the human adult brain. *Nat. Biotechnol.* **36**, 70–80 (2018).
45. McInnes, L., Healy, J. & Melville, J. UMAP: uniform manifold approximation and projection for dimension reduction. Preprint at *arXiv* <https://arxiv.org/abs/1802.03426> (2018).

Acknowledgements

We thank members of P. Bousso's laboratory for critical review of the manuscript. We thank the mouse facility and Technology Core of the Center for Translational Science (CRT) at Institut Pasteur for support in conducting the present study. The work was supported by Institut Pasteur, INSERM, a starting grant (Lymphocytecontact) and an advanced grant (ENLIGHTEN) from the European Research Council (to P. Bousso) and by the Bristol-Myers Squibb Foundation for Research in Immuno-Oncology.

Author contributions

R.T., I.M., M.C., F.L., Z.G., B.B. and C.C. conducted the experiments. R.T., I.M., M.C. and P. Bousso designed the experiments. I.A. and B.S. contributed to the sequencing analysis. R.T., I.M., M.C., P. Bost and P. Bousso analyzed the data and wrote the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at <https://doi.org/10.1038/s43018-020-0038-2>.

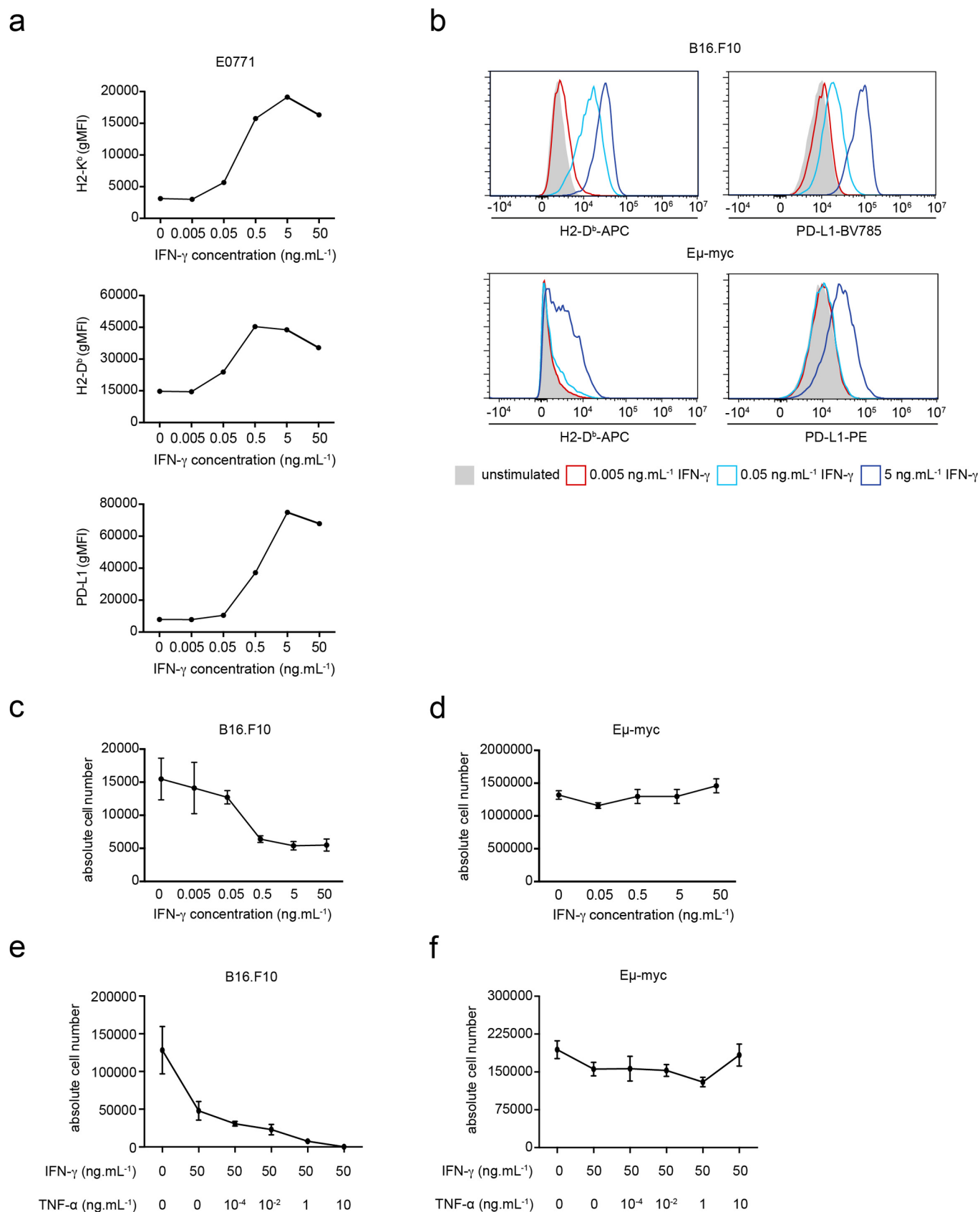
Supplementary information is available for this paper at <https://doi.org/10.1038/s43018-020-0038-2>.

Correspondence and requests for materials should be addressed to P.B.

Reprints and permissions information is available at www.nature.com/reprints.

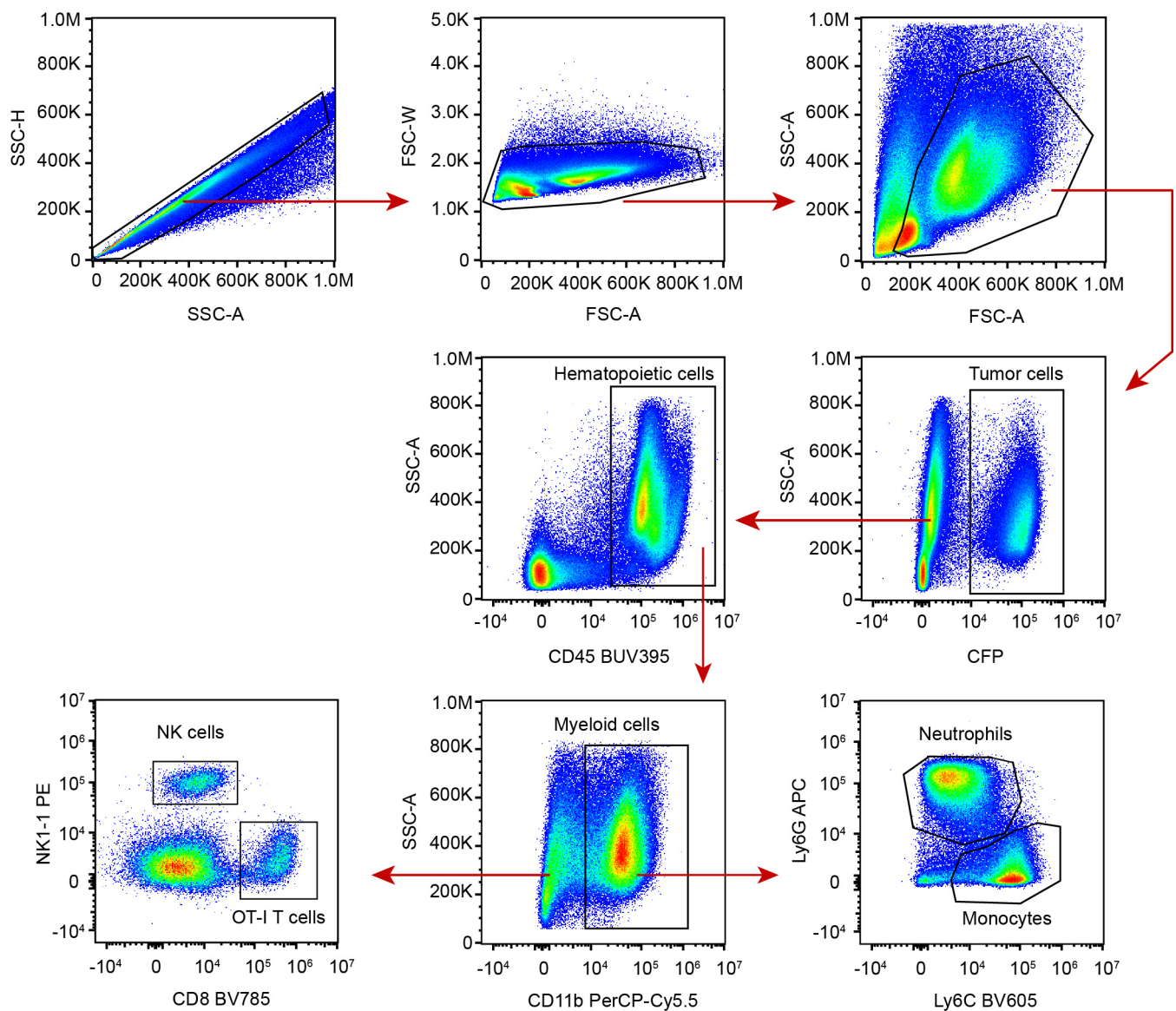
Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

© The Author(s), under exclusive licence to Springer Nature America, Inc. 2020

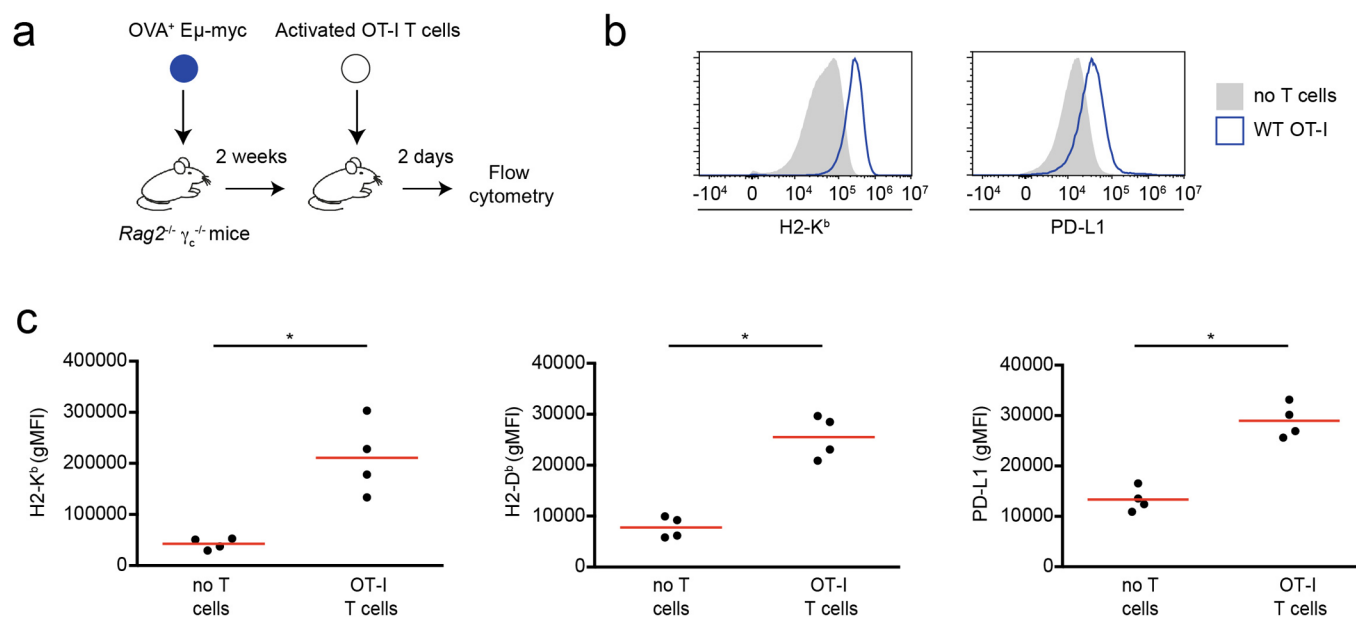


Extended Data Fig. 1 | See next page for caption.

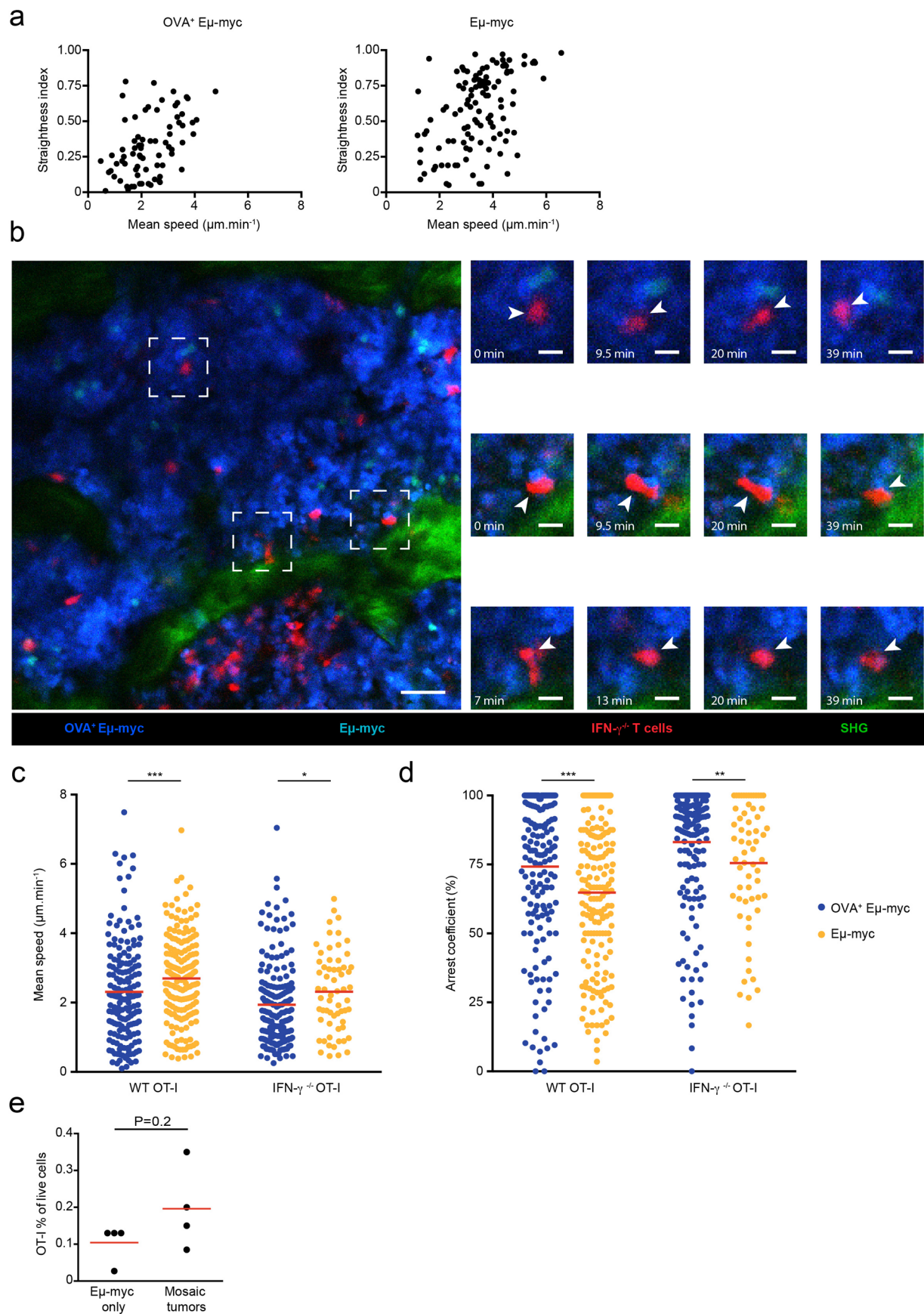
Extended Data Fig. 1 | Tumor cell responses to IFN- γ in vitro. **a-b**, Tumor cells respond to IFN- γ in an analogic manner in vitro. **a** E0771 mammary breast tumor cells were stimulated with indicated IFN- γ concentrations in vitro for 24h. H2-K^b (top), H2-D^b (middle) and PD-L1 (bottom) surface expression was then analyzed by flow cytometry. Representative of two independent experiments. **b**, Representative histograms of H2-D^b (left) and PD-L1 (right) surface expression in B16.F10 (top) or E μ -myc (bottom) cells, after treatment with indicated IFN- γ concentrations for 24h. Representative of three independent experiments. **c-f**, Impact of IFN- γ and TNF- α on tumor cell death. B16.F10 (**c,e**) or E μ -myc (**d,f**) cells were incubated with the indicated IFN- γ and/or TNF- α concentrations in vitro for 24h. Cells were then counted by flow cytometry. Representative of two independent experiments.



Extended Data Fig. 2 | Gating strategy for flow cytometry analysis of the tumor microenvironment. The figure depicts the gating strategies for identifying tumor cells, OT-I T cells, NK cells, monocytes and neutrophils present in the bone marrow of tumor bearing mice.

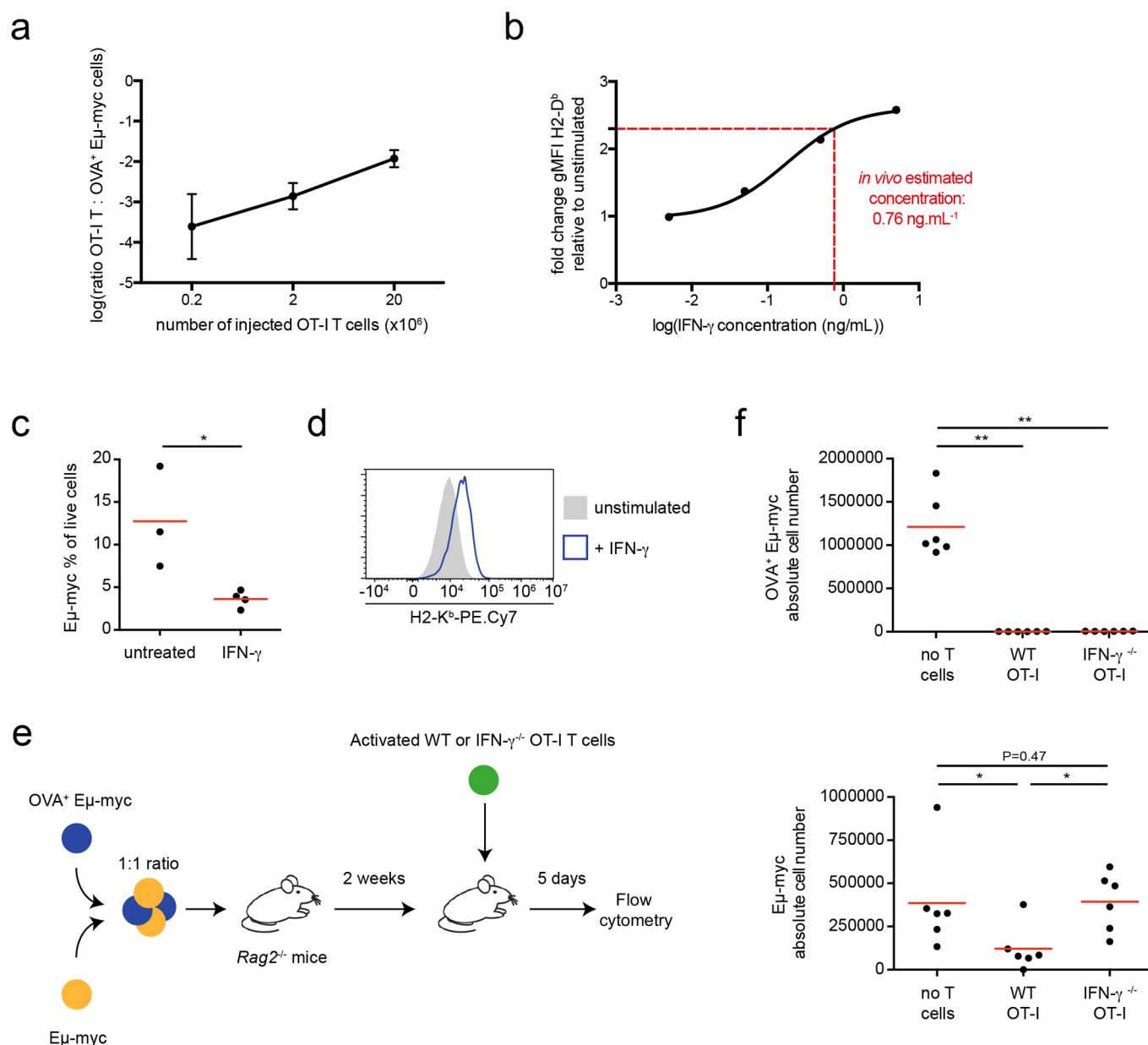


Extended Data Fig. 3 | NK cells are dispensable for tumor phenotypic changes upon T cell transfer. **a**, Experimental set-up. *Rag2^{-/-}γ_c^{-/-}* recipient mice were injected i.v. with OVA-expressing Eμ-myc B lymphoma cells. On day 12-13, in vitro activated OT-I CD8⁺ T cells were injected i.v. Two days later, the recipient bone marrow was harvested and analyzed by flow cytometry. **b**, Representative examples of histograms showing H2-K^b (left) and PD-L1 (right) surface expression on tumor cells isolated from the bone marrow of mice that were left untreated (filled grey) or injected with OT-I T cells (line, blue). **c**, H2-K^b (left), H2-D^b (middle) and PD-L1 (right) surface expression on tumor cells isolated from mice treated or not with OT-I T cells, as assessed by flow cytometry. Each dot represents one mouse with n=4 mice per group. Red lines indicate mean values. (* P<0.05, two-tailed Mann-Whitney U-test).



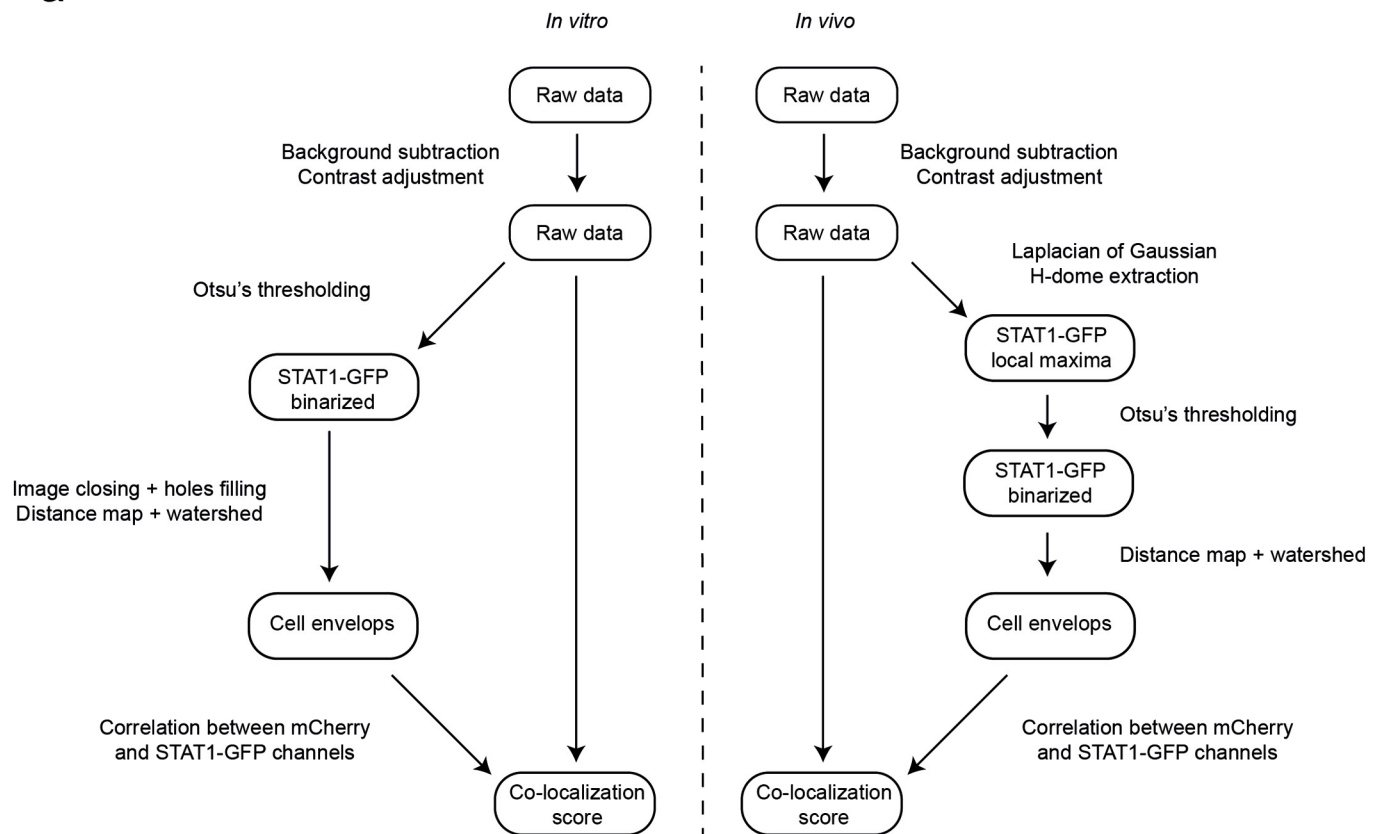
Extended Data Fig. 4 | See next page for caption.

Extended Data Fig. 4 | OT-I T cells arrest on antigen-positive tumor cells independently of IFN- γ . **a**, *Rag2*^{-/-} mice were injected with a 1:1 mixture of OVA-expressing and non-expressing E μ -myc B lymphoma cells, labeled with CFP and YFP, respectively. On day 12-13, mice were injected with activated GFP⁺ OT-I T cells. Two days later, intravital imaging of the bone marrow was performed. The graph shows the relationship between mean T cell velocity and straightness in the OVA⁺ (left) and OVA⁻ (right) tumor areas. Each dot represents one T cell track (OVA⁻ areas n= 103 tracks, OVA⁺ areas n=75 tracks). **b-d**, *Rag2*^{-/-} mice were injected with a 1:1 mixture of OVA-expressing and non-expressing E μ -myc B lymphoma cells, labeled with CFP and GFP, respectively. On day 12-13, mice were injected with activated WT or IFN- γ -deficient OT-I T cells transduced to express the mCherry fluorescent protein. Two days later, intravital imaging of the bone marrow was performed. **b**, Representative image of OVA⁺ (blue) tumor patches infiltrated with IFN- γ -deficient OT-I T cells (red). Scale bar: 50 μ m. Right. Time lapse images (corresponding to the dashed squares) showing IFN- γ -deficient OT-I T cells (pointed by arrows) forming stable contacts with OVA⁺ E μ -myc cells. Scale bar: 15 μ m. **c-d**, Both WT and IFN- γ -deficient OT-I T cells decelerate in OVA⁺ tumor areas. Graphs show mean velocities (**c**) and arrest coefficient (**d**), for individual WT or IFN- γ -deficient T cells in OVA⁺ (blue) and OVA⁻ (orange) tumor areas. Each dot represents one track. Red lines indicate mean values. Data shown in **b-d**, are representative of two independent experiments with n=3 mice per group. (* P<0.05; ** P<0.01; *** P<0.001; two-tailed Mann-Whitney U-test). **e**, OT-I T cells infiltrate antigen-negative tumors in the bone marrow. *Rag2*^{-/-} mice were injected with either E μ -myc alone or a 1:1 mixture of E μ -myc and OVA-expressing E μ -myc B lymphoma cells (labeled with different fluorescent proteins). On day 12-13, recipients were injected with OT-I T cells or left untreated. Two days later, the bone marrow of the mice was recovered and analyzed by flow cytometry. The graph shows that OT-I T cells can efficiently infiltrate tumors that contain only antigen-negative cells. Each dot represents one mouse with n=4 mice per group. Red lines indicate mean values.

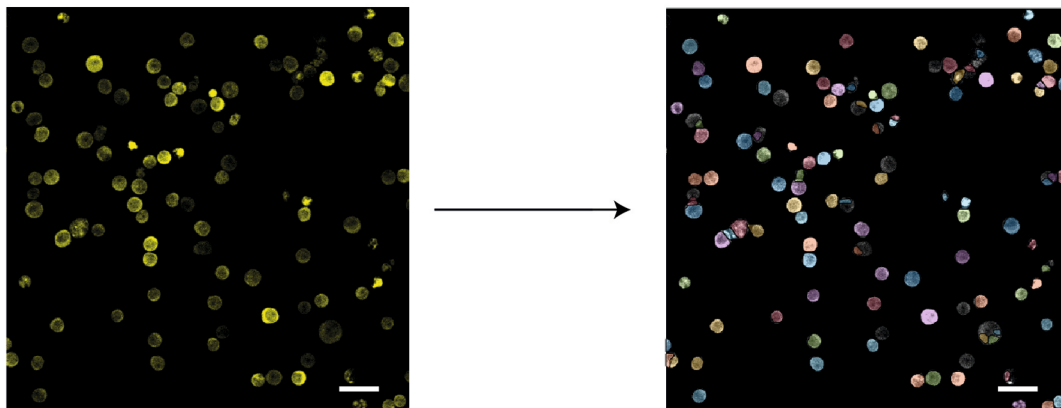


Extended Data Fig. 5 | T cell-derived IFN- γ can control bystander tumor cells. a-b, Estimating the intratumoral IFN- γ concentration upon T cell transfer. **a,** The ratio of OT-I T cells to OVA⁺Eμ-myc tumors in the bone marrow (measured at day 2) is graphed as a function of the total number of transferred T cells. Results are shown as mean $\pm$ SD with $n=5$ mice per group. **b,** The fold change in H2-D^b levels measured in Eμ-myc tumors *in vitro* is graphed as a function of IFN- γ concentration. The graph was used to infer the putative IFN- γ concentration *in vivo* (dashed red line) upon transfer of 20×10^6 T cells. **c-f,** Control of tumor burden by T cell-derived IFN- γ . **c-d,** Recipient female Rag2^{-/-} mice were injected with H-Y⁺ Eμ-myc B lymphoma cells. After 3 weeks, mice were either injected i.v. with recombinant IFN- γ (1 μ g) twice (one day apart) or left untreated. On day 2, tumor burden **c**, and phenotype **d**, in the bone marrow were assessed by flow cytometry. Each dot represents one mouse ($n=3$ and $n=4$ mice for the untreated and the treated group, respectively). Red lines indicate mean values. (* $P<0.05$, two-tailed unpaired t-test). **e-f,** Functional activity of intratumoral IFN- γ . **e,** Experimental set-up. Rag2^{-/-} mice were injected with a 1:1 mixture of Eμ-myc and OVA-expressing Eμ-myc B lymphoma cells (labeled with different fluorescent proteins). On day 12-13, recipients were injected with either WT or IFN- γ -deficient OT-I T cells or left untreated. Five days later, the bone marrow was recovered and analyzed by flow cytometry. **f,** Graphs show the residual numbers of antigen-positive (top) and antigen-negative (bottom) tumors following transfer of the indicated T cell populations. Each dot represents one mouse ($n=6$ mice per group). Red lines indicate mean values. (* $P<0.05$, ** $P<0.01$, two-tailed Mann-Whitney U-test). Representative of two independent experiments. Of note, small deviation from the initial ratio seen in the absence of T cells is most likely due to minor differences in tumor cell division time.

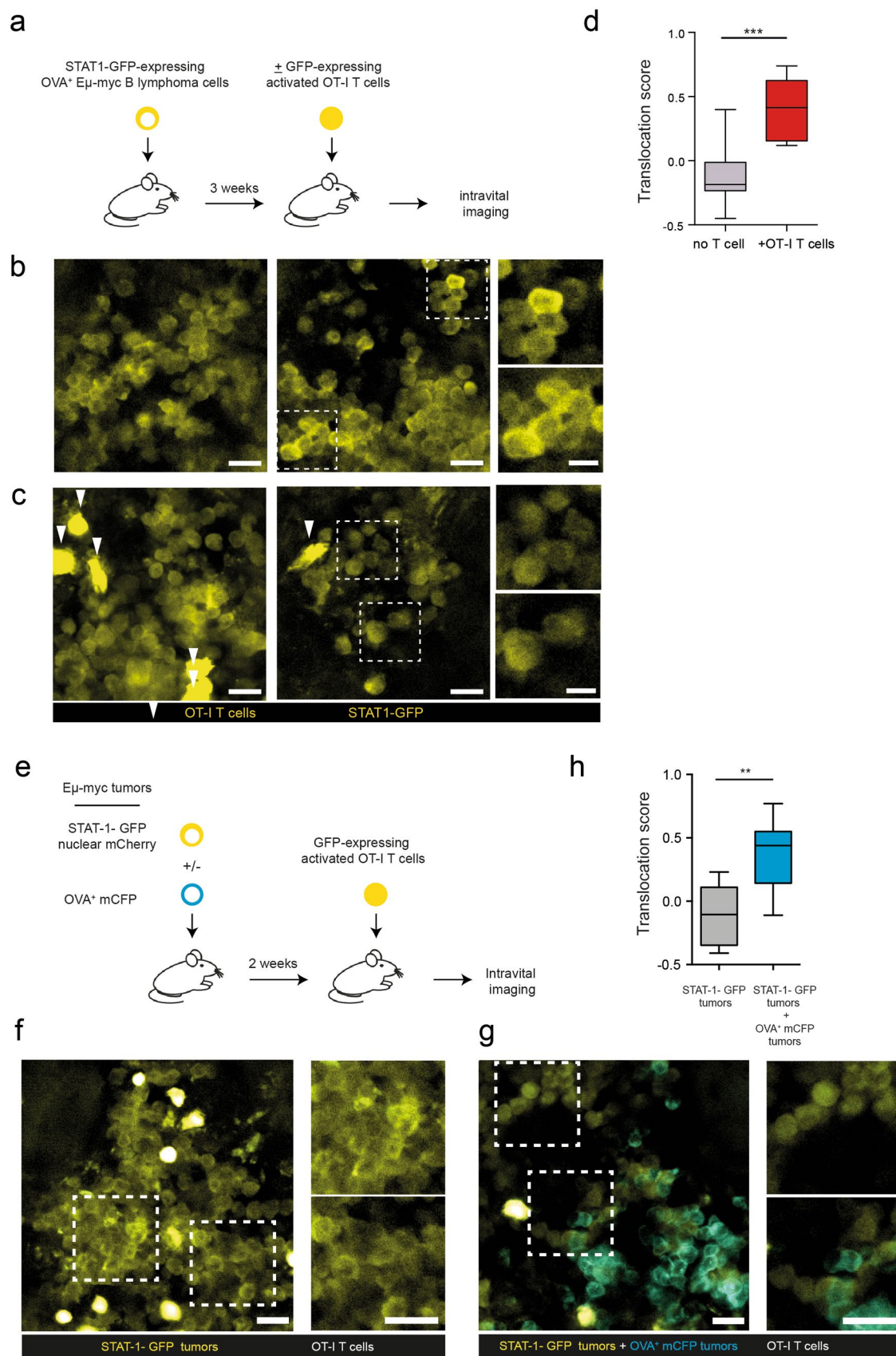
a



b

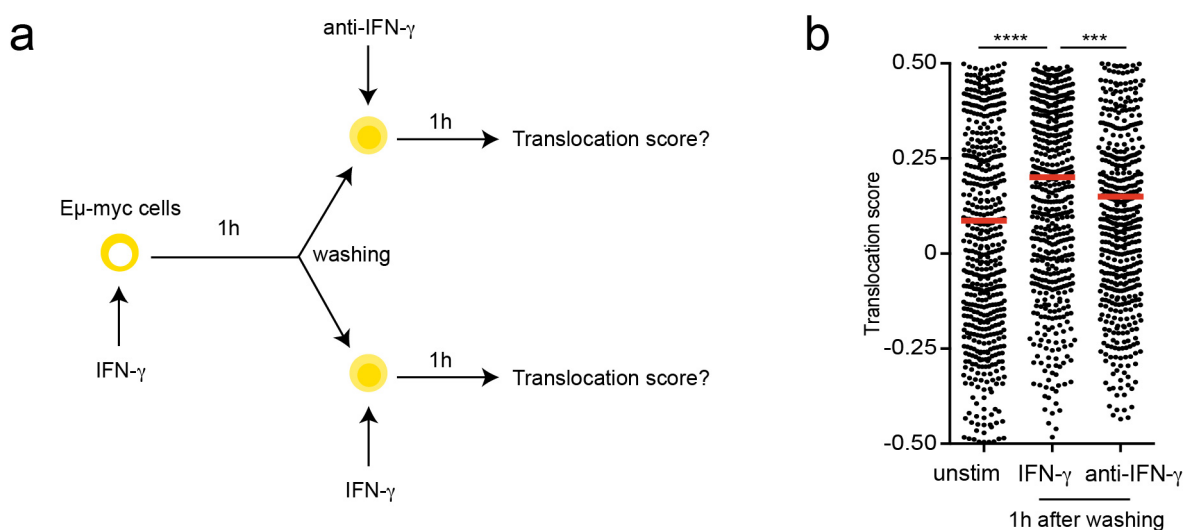


Extended Data Fig. 6 | Automated procedure for quantifying STAT1 translocation in tumor cells imaged *in vitro* or *in vivo*. **a**, The diagram recapitulates the various steps used for image processing and quantification (see methods for details). **b**, Representative example of cell segmentation from the STAT1-GFP signal in tumor cells. Scale bar: 50 μ m.

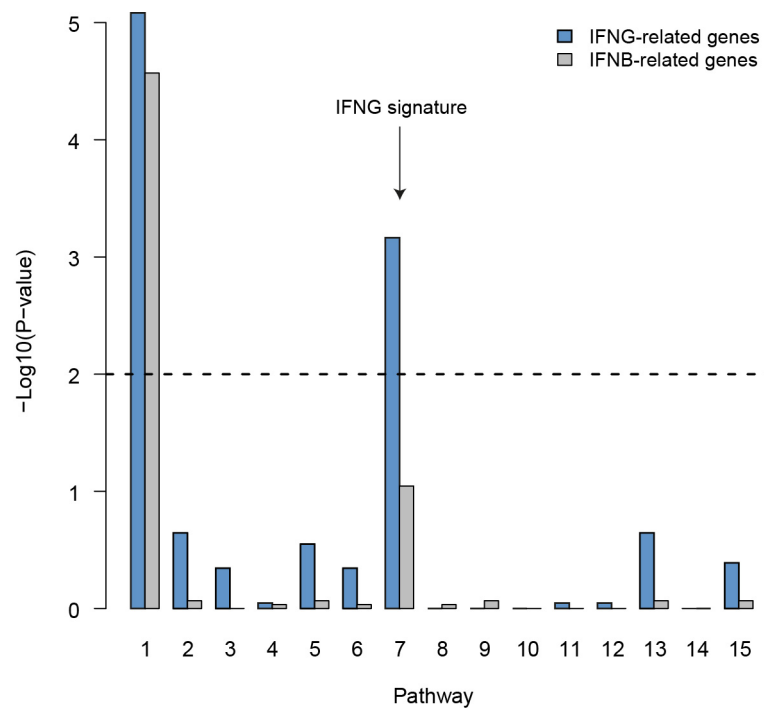


Extended Data Fig. 7 | See next page for caption.

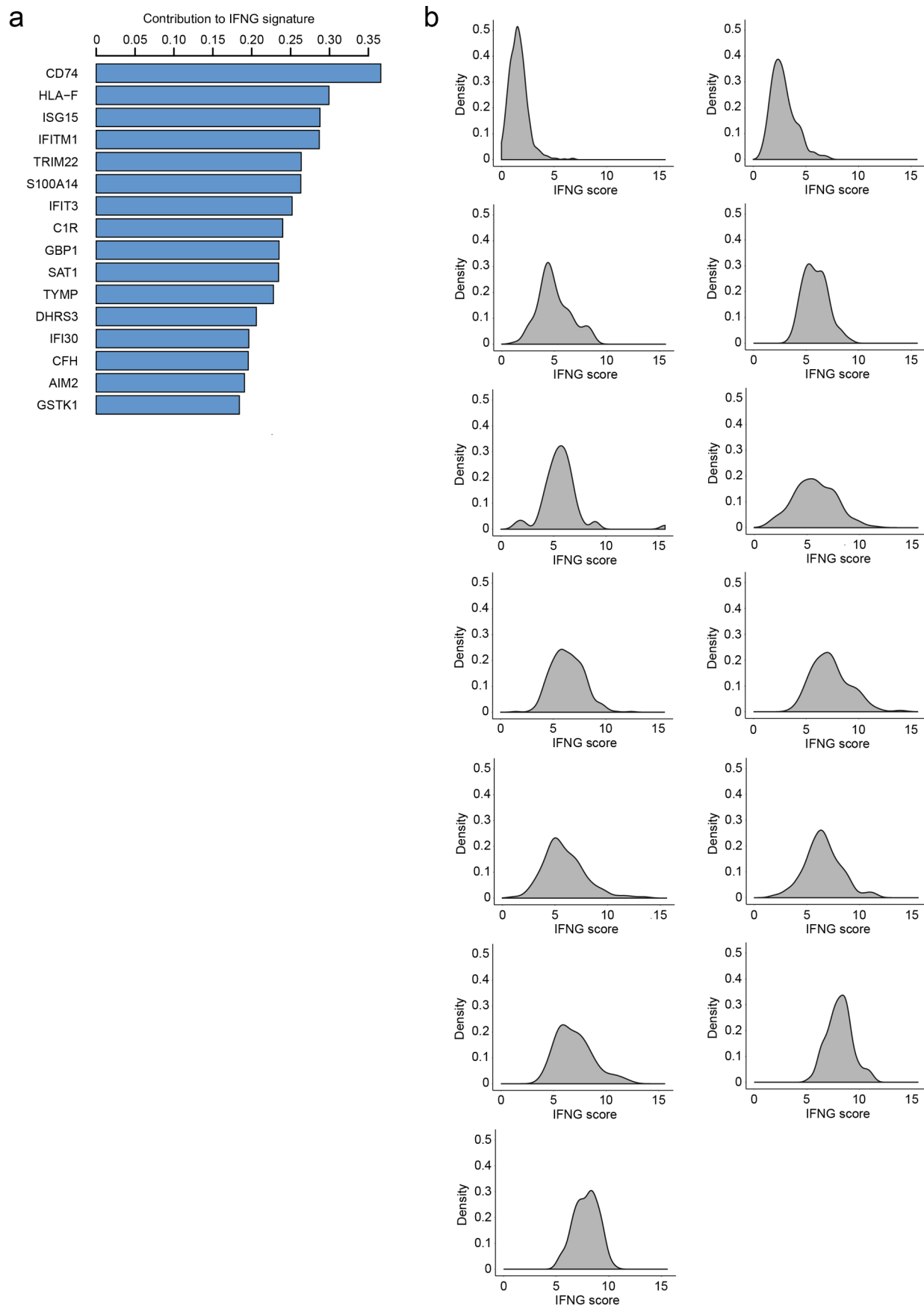
Extended Data Fig. 7 | Tumor-reactive CD8⁺ T cells mediate widespread and sustained STAT1 activity in the tumor microenvironment. **a**, Experimental set-up. Recipient *Rag2*^{-/-} γ_c ^{-/-} mice were injected with OVA⁺ E μ -myc B lymphoma cells expressing the STAT1-GFP reporter and a nuclear mCherry protein. After 3 weeks, activated CD8⁺ T cells bearing the OT-I TCR were injected i.v. One day later, recipients were subjected to intravital imaging of the bone marrow. **b**, STAT1-GFP is largely excluded from the nucleus in tumor cells developing in the absence of T cells. Representative two-photon images (scale bar: 20 μ m), highlighting two specific regions (insets, scale bar: 10 μ m). **c**, Detection of nuclear STAT1-GFP in T cell-infiltrated tumors. Representative two-photon images (scale bar: 20 μ m), highlighting two specific regions (insets, scale bar: 10 μ m). **d**, Translocation score was computed from two-photon images obtained in mice left untreated (no T cells) or transferred with OT-I T cells (n=10 cells per group, box plot showing the median, first and third quartile and min and max values, *** P<0.001, two-tailed Mann-Whitney U-test). Data in **b-d** are representative of two independent experiments. **e-h** STAT1 signaling is detected at distance from antigen-positive tumor cells. **e**, Experimental set-up. *Rag2*^{-/-} mice were injected with E μ -myc cells expressing the STAT1 reporter either alone or mixed at a 1:1 ratio with OVA⁺ E μ -myc (expressing mCFP). Two weeks later, all mice were adoptively transferred with activated OT-I T cells. After 2 days, mice were subjected to intravital imaging of the bone marrow. **f**, STAT1-GFP is largely excluded from the nucleus in tumor cells developing in the absence of antigen-positive tumors. Representative two-photon images (scale bar: 20 μ m), highlighting two specific regions. Scale bar, 20 μ m. **g**, Detection of STAT1 translocation in antigen-negative tumor cells developing in the presence of antigen-positive tumor cells. Representative two-photon images (scale bar: 20 μ m), highlighting two specific regions. Scale bar, 20 μ m. **h**, Translocation score was computed from two-photon images obtained in mice bearing STAT1-GFP-expressing tumor cells alone, or bearing mosaic tumors containing both STAT1-GFP and OVA⁺ tumor cells. (n=10 cells per group, box plot showing the median, first and third quartile and min and max values, ** P<0.01, two-tailed Mann-Whitney U-test). Data in **f-h** are representative of n=3 mice per group.



Extended Data Fig. 8 | Prolonged STAT1 translocation in tumor cells with sustained exposure to IFN- γ . **a**, Experimental set-up. E μ -myc cells expressing STAT1-GFP and nuclear mCherry were cultured for one hour in presence of recombinant IFN- γ . Cells were washed and either incubated with anti-IFN- γ mAb (to limit cytokine exposure), or re-incubated with IFN- γ (to prolong cytokine exposure). After an additional hour, cells were imaged. **b**, STAT1 translocation score was computed automatically for all individual cells (unstimulated $n=593$ cells. IFN- γ $n=583$, anti-IFN- γ $n=570$ cells). Each dot represents one cell. Red lines indicate mean values. (**** $P<0.0001$, *** $P<0.001$, two-tailed Mann-Whitney U-test).



Extended Data Fig. 9 | Determination of an *IFNG* signature in monocyte/macrophage cluster. 15 over-dispersed gene sets were determined for the monocyte/macrophage cluster as explained in materials and methods. Graph shows the contribution of *IFNG* or type I *IFN*-related genes for each gene sets. Arrow highlights a gene signature that is specific of IFN- γ signaling (hereafter called *IFNG* signature).



Extended Data Fig. 10 | Distribution of IFN- γ signature in tumor cells from head and neck squamous cell carcinoma patients. **a, Gene contribution to the IFNG signature in the tumor cells. **b**, Distribution of IFNG signature in tumor cells from 13 different patients with head and neck squamous cell carcinoma patients. Although the mean of the distribution varies from patient to patient, a relatively uniform distribution is observed in most patients.**

Reporting Summary

Nature Research wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Research policies, see [Authors & Referees](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a Confirmed

- ☐ ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement
- ☐ ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
- ☐ ☒ The statistical test(s) used AND whether they are one- or two-sided
Only common tests should be described solely by name; describe more complex techniques in the Methods section.
- ☒ ☐ A description of all covariates tested
- ☐ ☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
- ☐ ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
- ☐ ☒ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
Give P values as exact values whenever suitable.
- ☒ ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
- ☒ ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
- ☐ ☒ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection

For FACS experiments, samples were acquired using either BD FACSDiva software (BD Biosciences) or CytExpert 2.3 software (Beckman Coulter).
For microscopy experiments, movies were acquired using FV31S-SW software (Olympus)

Data analysis

Flow cytometry data were analyzed using FlowJo v10.4.1 (Tree Star). Microscopy movies were created from raw data and analyzed using Fiji software (Image J v1.50i). Alternatively, analysis was performed on Imaris 7.4.2 (Bitplane) or Matlab2019b(R). Data were represented using GraphPad Prism v6.
For RNAseq data, Bowtie2 was used to determine rRNA abundance. Quality control was performed using Picard and RSeQC. Reads were aligned using STAR (version 2.5.2a) and quantified with featureCounts (version 1.4.3-p1). Further analysis was performed using R (version 3.4.3) with the DESeq2 package (version 1.18.1). Enrichment of specific transcription factors motifs was analyzed using the iRegulon Cytoscape plugin.
For single-cell RNAseq data, data were analyzed using R once again, with the GEOquery (version 2.48.0), Pagoda2, igraph packages.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors/reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

Already published Single-cell RNAseq data were downloaded from GEO, accession number GSE123139

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No sample size calculation was performed. Sample sizes were determined based on previous experiments in the laboratory, in order to get statistically significant results and limit mice number at the same time.
Data exclusions	For microscopy experiments, mice with in which the fluorescent signal to noise ratio was too low were not analyzed. The criteria for exclusion was the inability to discriminate unambiguously individual cell countours.
Replication	Experiments were repeated several times, as indicated in figure legends. All repetitions of the experiments yielded similar results. Mice were housed in the same facility, in the same room and were age-matched as best as possible in every experiments and between different experiments. At the time of injection, tumor cells were less than one month in culture and all fluorescent cell lines were prepared from the same original uncolored line.
Randomization	Age- and sex-matched mice were used. For experiments using B16.F10 melanoma cells, different groups of mice were randomly attributed to different experimental conditions. For experiments using B cell lymphoma, different groups of mice were attributed to different experimental conditions according to the percentage of tumor cells in the blood, in order to have comparable groups in terms of tumor growth before T cells injection
Blinding	The experimentators were not blinded to experimental conditions during experiments of experiments analysis. Experiments and analyses were often performed by the same person.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

Methods

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology
☐	☒ Animals and other organisms
☒	☐ Human research participants
☒	☐ Clinical data

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

The following antibodies were used:

-anti-mouse PD-L1-biotin; clone 10F.9G2, BioLegend # 124305, lot #B220794, dilution 1/100
 -anti-mouse PD-L1-BV785; clone 10F.9G2, BioLegend # 124331, lot #B253998, dilution 1/100
 -anti-mouse H2-Kb-PE-Cy7, clone AF6-88.5, BioLegend # 116519, lot #B248766, dilution 1/100
 -anti-mouse H2-Kb-BUV395, clone AF6-88.5, BD Biosciences # 742864, lot #8227669, dilution 1/100
 -anti-mouse H2-Db-APC, clone KH95, BioLegend # 111514, lot #B242441, dilution 1/100
 -anti-mouse CD8-BUV395, clone 53-6.7, BD Biosciences # 563786, lot #8072932, dilution 1/200
 -anti-mouse CD8-BV786, clone 53-6.7, BD Biosciences #563332, dilution 1/200
 -anti-mouse IFN-g, purified; clone XMGI.2, BioXcell #BE0055, lot #630017J1. Used at 50 µg/mL for in vitro blocking purposes
 -anti-mouse IFN-g-APC, clone XMGI.2, eBioscience #51-7311-82, lot #E08729-1131, dilution 1/100
 -anti-mouse/human CD11b, clone M1/70, BioLegend #101228, lot #B260459, dilution 1/100
 -anti-mouse NK1.1-PE, clone PK136, eBioscience #12-5941-83, lot #E030343, dilution 1/100
 -anti-mouse Ly6G-APC, clone 1A8, BioLegend #127610, lot #B255839, dilution 1/100
 -anti-mouse Ly6C-BV605, clone HK1.4, BioLegend #128036, lot #B219484, dilution 1/100

-anti-mouse CD45-BUV395, clone 30-F11, BD Biosciences #564279, lot #8262745, dilution 1/100
 -anti-mouse CD16/32, purified, clone 93, BioLegend #101302, lot #B225480, dilution 1/100
 -streptavidin-PE, eBiosciences #12-4317-87, lot #E034484, dilution 1/400

Validation

All antibodies were validated by the corresponding manufacturers. For each experiments, unstained controls were added to validate the positivity of the stainings. Relevant information on antibodies validation can be found on the manufacturer's website for each antibody.

Eukaryotic cell lines

Policy information about [cell lines](#)

Cell line source(s)

The Eμ-myc cell line was derived in our laboratory from a male Eμ-myc transgenic mouse. Eμ-myc mice harbor a transgene with the c-myc gene under the IgH promoter and develop spontaneous B lymphoma.
 HEK cells were purchased at American Type Culture Collection.
 B16.F10 melanoma cells were a gift from Guy Shakhar (Weizmann Institute, Rehovot, Israel).

Authentication

HEK and B16.F10 cells were previously authenticated by ATCC (see vendor website). The phenotype of the cell lines used in the study was checked after thawing of the cells.

Mycoplasma contamination

HEK and B16.F10 cells were tested negative for mycoplasma contamination by PCR. Eμ-myc cells were also tested negative at the exception of YFP+ cells which were not tested.

Commonly misidentified lines (See [ICLAC](#) register)

No commonly misidentified cell lines were used.

Animals and other organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research

Laboratory animals

C57BL/6J-Rag2tm (Rag2-deficient mice), C57BL/6J-Tg(UBC-GFP)30Scha Tg(TcraTcrb)1100Mjb Rag1tm1Bal (Rag1-deficient; OT-I TCR transgenic, GFP-expressing mice), C57BL/6J-Tg(Act-CFP) (expressing CFP under the actin promoter), C57BL/6J-Tg (TCR Matahari) Rag2tm1Fwa (Rag2-deficient, Matahari TCR transgenic) and C57BL/6J-Rag2tm1Fwa IL2rgtm1 Cgn (Rag2 and g common chain-deficient) were housed and bred in our facility.

For experiments, involving OT-I cells transfer, age- and sex-matched male or female mice were used. For experiments involving Matahari T cells transfer, only age-matched female mice were used. Mice were 6 to 25 weeks old at the time of the experiments and all experiments were performed in agreement with relevant Institut Pasteur internal guidelines and regulations (CETEA committee, protocol number 170038)

Wild animals

The study did not involve wild animals

Field-collected samples

The study did not involve field-collected samples

Ethics oversight

All procedures were approved by the Institut Pasteur Committee on Animal Welfare (CETEA) under protocol code 170038, as well as the French Superior Teaching and Research Ministry

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Flow Cytometry

Plots

Confirm that:

- ☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- ☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- ☒ All plots are contour plots with outliers or pseudocolor plots.
- ☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation

For experiments using B16.F10 melanoma in vivo, whole tumors were recovered and cut into small parts, then digested with Collagenase and DNase I for 30min at 37°C. Digested tumors were then further crushed on a cell strainer and tumor cells were recovered before antibody stainings and flow cytometry analysis.

For experiments using Eμ-myc lymphoma in vivo, tumor growth was monitored by measuring tumor cell percentage in the blood. To do so, blood was recovered in EDTA and red blood cells were lysed using ACK solution before analysis at the cytometer. Once tumors were established, bone marrow cells were recovered by flushing one tibia and one femur with FACS buffer (PBS supplemented with 0.5% FCS) through a cell strainer. Single-cell suspensions were then recovered and stained before analysis by flow cytometry

Instrument	A majority of the data was acquired using a Cytoflex LX (serial number BA42021). Some data were acquired using a BD Canto II
Software	BD FACS Diva was used for BD Canto II-acquired Data. For cytoflex-acquired data, CytExpert version 2.3 was used
Cell population abundance	No cell sorting was performed during experiments
Gating strategy	For experiments using B16.F10 melanoma cells, cell doublets were gated out based on SSC-A/SSC-H followed by an additional gating based on FSC-A/FSC-W. Live cells were isolated by FSC-A/SSC-A gating. Injected OT-I T cells were distinguished based on their GFP expression. B16.F10 cells expressing OVA or not were then gated on based on their expression of TdTomato and membrane CFP respectively. A back-gating on FSC-A/SSC-A was used to confirm the discrimination of B16.F10 cells. For experiments using Eμ-myc B cell lymphoma in vivo, cell doublets were excluded as described above. CFP and GFP were used to detect Eμ-myc expressing OVA or not respectively. CD8 staining enabled further distinction of GFP+ OT-I T cells from GFP+ tumor cells. In a distinct panel, CD45 enabled distinction of hematopoietic cells. OVA- Eμ-myc cells and OT-I T cells were gated out using GFP and CD8 expression as above. Likewise, OVA+ Eμ-myc cells were gated out by their CFP expression. Among remaining cells, myeloid cells were distinguished by their CD11b expression and further divided into monocytes (Ly6C high Ly6G-) and neutrophils (Ly6C+ Ly6G+). Finally, NK cells were identified as CD45+ CD11b- NK1.1+. A figure describing the gating strategy that was used to identify tumor, NK, OT-I T cells, as well as monocytes and neutrophils, is provided in the manuscript.

☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.